63.1 1. A local family with the correct hypercubic transformation rule

Let φ be the massless free scalar field in four dimensions. On separated Euclidean tests its covariance is

$$C(x) = \frac{1}{4\pi^2|x|^2}. \tag{TC1}$$

Define a symmetric trace-free array of local derivative fields by

$$B_{\mu\nu}(x) = \begin{cases} (\partial_\mu^2 - \frac{1}{4}\Delta)\varphi(x), & \mu = \nu, \\ 0, & \mu \neq \nu. \end{cases} \tag{TC2}$$

The diagonal expression has no summation over μ. Derivatives are defined by transferring them to test functions. Thus this is an actual jointly defined field family, not just a collection of proposed two-point functions.

Signed permutations of the coordinate axes permute the diagonal second derivatives in precisely the way they permute diagonal tensor components. They preserve the zero off-diagonal components. Hence (TC2) transforms as a symmetric rank-two tensor under the full hypercubic group.

Every coordinate mirror and every diagonal mirror in that group has reflection positivity with the corresponding tensor reflection matrix. Indeed replacing each $B(f)$ in a positive-half polynomial by the derivative expression (TC2) gives a positive-half polynomial in the free scalar field. Differentiation preserves test support and commutes with those reflections in exactly the tensor rule just described. Its reflection form is therefore the original nonnegative free-field form. Translations, locality, the forward spectral cone, and finite energy bounds are inherited from the same free field.

63.2 2. The continuum tensor law fails

A rotation by $\pi/4$ in the $1, 2$ plane takes a diagonal trace-free tensor difference into an off-diagonal component. If (TC2) obeyed the full rank-two covariance law, the identically zero field B_{12} would

consequently imply

$$B_{11} - B_{22} = 0.$$

But this difference is $(\partial_1^2 - \partial_2^2)\varphi$, and its separated two-point function is not zero. For example, at $x = te_4$,

$$\begin{aligned} \big\langle (B_{11} - B_{22})(te_4)(B_{11} - B_{22})(0) \big\rangle_E &= (\partial_1^2 - \partial_2^2)^2 C(te_4) \\ &= \frac{8}{\pi^2 t^6} > 0. \end{aligned} \tag{TC3}$$

The coefficient follows by expanding $(t^2 + x_1^2 + x_2^2)^{-1}$: its fourth derivatives give $24 - 2 \cdot 8 + 24 = 32$, followed by the factor $1/(4\pi^2)$. Thus all the stated lattice symmetries and positivity properties coexist with a failure of the natural continuum tensor law.

This example is not a counterexample to the scalar angular theorem. Its individual diagonal entries are not scalars under exchanges of coordinate axes. Applying a scalar quarter-turn identity to each component would already use a false premise.

63.3 3. Leading ultraviolet matching does not remove this ambiguity

The covariance of B scales as $|x|^{-6}$. It is therefore subleading to a dimension-four covariance $|x|^{-8}/\log^2(1/|x|)$. In particular

$$|x|^8 \log^2(1/|x|)\, \langle B(x)B(0)\rangle_E \longrightarrow 0. \tag{TC4}$$

On fixed-amplitude profiles $f_d(x) = f(x/d)$, this same contribution is $O(d^2)$.

More generally, suppose a candidate covariant tensor family T with that leading two-point law has been constructed on some Hilbert space. On its tensor product with an independent free scalar theory, form

$$T'_{\mu\nu} = T_{\mu\nu} + \mu B_{\mu\nu}, \qquad \mu \neq 0. \tag{TC5}$$

The fixed dimensionful factor μ makes the engineering units match. The leading tensor ultraviolet covariance is unchanged by (TC4). Since B is jointly Gaussian and independent, every connected correlation of order at least three in the T' family is exactly the corresponding connected correlation in T. Mixed connected correlations with any fields from the original theory are unchanged as well, unless both insertions in a two-point function are replaced by T'.

The joint hierarchy still has translation invariance, locality, positive energy, the same hypercubic tensor covariance, and the corresponding lattice mirror reflection positivity. Nevertheless the natural full tensor covariance fails: subtracting the covariance identity for T from the proposed identity for T' would impose that identity on the nonzero independent field B, contradicting (TC3).

This is a logical test of the proposed reconstruction implications. It is not asserted to be a Wilson continuum limit, and no existence of the hypothetical base family T is being proved by (TC5). It shows why the original Wilson observable correspondence and its allowed lower-dimensional mixing must enter a tensor argument. Leading short-distance matching, even together with all the higher connected correlations of a proposed family, does not by itself identify how each additional field transforms.

63.4 4. The requirement for the present construction

The 21 curvature-bilinear and 56 charge-odd cubic sectors have specified finite lattice transformation rules. To obtain their continuum tensor laws one needs either a genuine tensor rotation theorem

using the completed physical mixed bounds, or a proved operator correspondence that rules out admixtures of the kind above and fixes the rotation action on the dimension-filtered family. The scalar action-density reconstruction alone supplies neither statement. This remains a separate task even if the physical multi-insertion estimates for the two finite sectors are completed.

64.1 1. The finite-cutoff identity, including the measure derivative

On a finite product of compact link groups, let $L_{b,A}$ be the left-invariant derivative in color direction A on bond b. For a smooth vector field

$$V = \sum_{b,A} v_b^A(U) L_{b,A}, \qquad \operatorname{div}_H V = \sum_{b,A} L_{b,A} v_b^A,$$

Haar integration by parts in the Wilson probability measure gives

$$\boxed{\langle VP \rangle_a = \langle P\,[VS_a - \operatorname{div}_H V] \rangle_a.} \tag{WT1}$$

Indeed integrate $L_{b,A}(v_b^A P e^{-S_a})$ over that compact group factor and sum. No gauge fixing, Gaussian approximation, continuum field, or boundary condition at infinity is needed. For a specially chosen divergence-free lattice translation vector field the divergence term vanishes. It must otherwise be retained.

Choose a local covariant translation variation $V_{a,\alpha}$ with smooth compact vector test α. Once a candidate renormalized stress tensor $T^R_{a,\mu\nu}$ and a normalization of this variation have been fixed, define its residual by the exact identity

$$Z_{\delta,a}[V_{a,\alpha} S_a - \operatorname{div}_H V_{a,\alpha}] = a^4 \sum_x (\nabla^a_\mu \alpha_\nu)(x) T^R_{a,\mu\nu}(x) + \mathcal{R}_a(\alpha). \tag{WT2}$$

Thus, for every finite-cutoff probe P_a,

$$Z_{\delta,a} \langle V_{a,\alpha} P_a \rangle_a = \left\langle P_a\, a^4 \sum_x (\nabla^a_\mu \alpha_\nu) T^R_{a,\mu\nu} \right\rangle_a + \langle P_a \mathcal{R}_a(\alpha) \rangle_a. \tag{WT3}$$

These identities are exact definitions and integration by parts. They do not assert that the residual is small.

64.2 2. A quantitative condition sufficient for this limit passage

Fix a finite admitted operator family and separated tests of scale and mutual separation at least $h > 0$. Let P_a be any product of m of its already normalized insertions. A useful residual estimate would be

$$|\langle P_a \mathcal{R}_a(\alpha) \rangle_a| \le C^{m+1} ((m+1)!)^p h^{-b(m+1)} \left(\frac{a}{h}\right)^\eta (1 + x_a)^q \|\alpha\|_{C^s} \prod_{i=1}^m \|f_i\|_{C^s}, \qquad \eta > 0, \tag{WT4}$$

with constants independent of the fine cutoff, the compatible volume, and the chosen probe. A finite sum of analogous powers would also suffice. Since $x_a = O(\log(1/a))$, the right side tends to zero for each fixed collection of tests.

One also needs the physical variation comparison

$$Z_{\delta,a}\langle V_{a,\alpha} P_a\rangle_a \longrightarrow \langle \delta_\alpha P\rangle, \tag{WT5}$$

where δ_α is the specified infinitesimal action on the continuum operator family, including the tensor-index terms when those are required. Together with convergence of the stress insertions, (WT3)–(WT5) give the corresponding continuum Ward identity. Distributional extensions and possible contact terms must be specified in the same scheme on both sides.

The inequalities are written for completed physical correlations. A bound on a positive reference coarse density, or a classical statement $\mathcal{R}_a = O(a^2)$ on smooth configurations, does not imply (WT4). Near an inserted composite, inverse powers of the cutoff can multiply that nominal a^2. This is the same near-mark issue isolated by the completed-coefficient analysis.

Fitting a finite number of stress normalization coefficients to an equal number of calibration probes does not prove (WT4). Such a fit only annihilates a finite number of values of the residual functional. A uniform bound on its remaining component is required to infer the Ward identity for every probe.

65.1 1. A precise local hypothesis

Write $d = 4$, and let $S_{n,a}$ be distributions at finite cutoff, or multilinear correlation functionals with their usual distributional extensions. Fix integers $s \geq 0$ and real constants $b, p \geq 0$. Assume that for every $0 < h \leq 1$, every collection of cubes Q_i of side comparable to h, whose fixed enlargements remain separated by at least ch, and all smooth f_i supported in those enlargements,

$$|S_{n,a}(f_1 \otimes \cdots \otimes f_n)| \leq AB^n (n!)^p h^{-bn} \prod_{i=1}^{n} \|f_i(c_i + h\,\cdot)\|_{C^s}. \tag{D1}$$

Here c_i are the cube centers. Constants are independent of the cutoff, positions, and the number of cubes. A fixed multiplicative change in cube size or separation is absorbed into B. Translation invariance is one way to obtain the position independence, but it is not used in the proof below. A polynomial position weight can be added at the expense of the final Schwartz seminorm.

The bound concerns the full correlation, not just its connected part. Connected bounds can be converted first by summing set partitions; the Bell-number cost is bounded by $n^n \leq e^n n!$, increasing p by one when the other constants have the corresponding multiplicative form.

For $n = 1$, assume the same local bound or choose the vacuum subtraction so that the one-point function vanishes. No coincidence diagonal is present in that case.

65.2 2. A local nuclear expansion with order linear in n

Put

$$q = 2 \left\lceil \frac{s + d + 1}{2} \right\rceil.$$

After translation and dilation, consider a smooth function F supported strictly inside the product of n fixed unit cubes. Periodize it on slightly larger cubes, and choose one fixed cutoff χ that is one on every projection of its support. Its Fourier series has the form

$$F(y_1, \ldots, y_n) = \sum_{k_1, \ldots, k_n \in \mathbb{Z}^d} c_{\boldsymbol{k}} \prod_i \chi(y_i) e^{i\pi k_i \cdot y_i / C}. \tag{D2}$$

Integrate by parts with $\prod_i (1 - \Delta_{y_i})^{q/2}$. The number of derivative terms is a fixed constant to the nth power. Consequently

$$|c_{\boldsymbol{k}}| \leq C^n \|F\|_{C^{qn}} \prod_i (1 + |k_i|^2)^{-q/2}. \tag{D3}$$

The scaled C^s norm of the ith factor in (D2) is at most $C(1 + |k_i|)^s$. Since $q > s + d$, the sum of their products times $|c_{\boldsymbol{k}}|$ is bounded by $C^n \|F\|_{C^{qn}}$. Thus the series converges in the projective tensor norm controlled by (D1). This proves the local joint-test bound

$$|S_{n,a}(F)| \le A(BC)^n (n!)^p h^{-bn} \max_{|\beta|\le qn} h^{|\beta|} \|\partial^\beta F\|_\infty. \tag{D4}$$

No pointwise correlation kernel was used. At a fixed cutoff one can justify equality with the original distribution by convergence of the smooth Fourier series in every prescribed finite derivative order; alternatively, (D2) defines the unique continuous extension on this product chart. Overlapping charts agree on finite sums of tensors and then by the same density argument.

65.3 3. A separation partition with controlled dependence on n

Choose a smooth radial function u with values in $[0,1]$, equal to zero for $|x| \le 1$ and to one for $|x| \ge 2$, monotonically increasing in $|x|$. It can be chosen in a fixed Gevrey class, for example with

$$\|\partial^\beta u\|_\infty \le C^{|\beta|+1} (|\beta|!)^2. \tag{D5}$$

For $j \ge 0$, set

$$U_j(\boldsymbol{x}) = \prod_{i<l} u(2^j (x_i - x_l)), \qquad \eta_0 = U_0, \qquad \eta_j = U_j - U_{j-1} \quad (j \ge 1). \tag{D6}$$

On the configuration space with all points distinct, the nonnegative functions η_j sum to one. For $j \ge 1$, a nonzero η_j requires

$$2^{-j} < \min_{i<l} |x_i - x_l| < 2^{2-j}. \tag{D7}$$

There are at most $n^2/2$ factors in (D6). Leibniz's rule and (D5) give, for derivative order k,

$$\|\partial^\beta \eta_j\|_\infty \le (Cn^2)^k C\,(k!)^2\, 2^{jk}, \qquad |\beta| = k. \tag{D8}$$

The bound is deliberately crude. When $k \le qn$, its dependence on n is at most $C^n (n!)^{c_q}$ for a fixed c_q. In particular it is not an uncontrolled C^{n^2} loss. This is the reason for using an explicit product partition instead of a dimension-dependent Whitney normalization whose derivative constants have not been tracked.

At level j, use a product partition of unity by cubes of side h_j/K, where $h_j = 2^{-j}$ and K is a fixed sufficiently large constant. A one-variable partition is obtained by convolving unit interval indicators with a fixed compact smooth bump, then tensoring over the four spatial coordinates. Tensoring it over the n points introduces only C^n overlap and Gevrey derivative costs of the same kind as (D8).

If the product of a cube cutoff with η_j is nonzero, its projected cubes and the slightly larger supports used for the Fourier cutoffs remain pairwise separated by a fixed multiple of h_j. Indeed a point in that product has separation at least h_j, while each projection moves by at most Ch_j/K. Thus (D4) applies to every nonzero chart. Cubes whose product does not meet that separation shell make no contribution.

65.4 4. Flatness pays the entire short-distance singularity

Let F be a Schwartz function flat on every coincidence diagonal, meaning that every derivative vanishes there. On the support of η_j, choose a nearest pair and project it onto its coincidence diagonal. The

displacement has only eight possibly nonzero coordinates and length at most Ch_j. For every $|\beta| \le qn$, Taylor's formula applied to $\partial^\beta F$ gives

$$|\partial^\beta F(\boldsymbol{x})| \le \frac{(Ch_j)^M}{M!} \max_{|\gamma|\le M+|\beta|} \sup_{|y-\boldsymbol{x}|\le Ch_j} |\partial^\gamma F(y)|. \tag{D9}$$

The constant in the directional expansion is independent of n because only that pair of points moves. The nearest pair can depend on $\boldsymbol{x}$; this is a pointwise estimate and no derivative of the choice is taken.

Take

$$M = \lceil (b+d)n \rceil + 2. \tag{D10}$$

For a fixed integer $w > d$, define the product-weight seminorm

$$P_{r,w,n}(F) = \sup_{|\gamma|\le r,\, \boldsymbol{x}} \left[\prod_i (1+|x_i|)^w \right] |\partial^\gamma F(\boldsymbol{x})|. \tag{D11}$$

The displacement in (D9) changes only two position weights, by a fixed factor, since $h_j \le 1$. Combining (D4), the cutoff derivative bounds, and (D9) shows that a chart at scale h_j, with centers $c_1, \dots, c_n$, contributes at most

$$AC^n (n!)^{p'} h_j^{M-bn} P_{M+qn,w,n}(F) \prod_i (1+|c_i|)^{-w}. \tag{D12}$$

Here p' is fixed independently of n, j, a. The factors $n^{O(n)}$, $((qn)!)^2$, and the finite multi-index Leibniz sums are all absorbed into $C^n (n!)^{p'-p}$. The favorable $1/M!$ may simply be discarded.

Summing the position grids gives

$$\sum_{c_1,\dots,c_n} \prod_i (1+|c_i|)^{-w} \le C^n h_j^{-dn}. \tag{D13}$$

By (D10), the remaining scale factor is at most h_j^2, and its dyadic sum is finite. The $j = 0$ contribution is handled by the same cube and Fourier argument at scale one, using Schwartz decay without Taylor expansion. Thus

$$|S_{n,a}(F)| \le AC^n (n!)^{p'} P_{M+qn,w,n}(F). \tag{D14}$$

Choose one fixed integer $k > q + b + d + w + 3$. Then $M + qn \le kn$ for $n \ge 2$, and

$$\prod_i (1+|x_i|)^w \le (1+|\boldsymbol{x}|)^{wn} \le (1+|\boldsymbol{x}|)^{kn}.$$

Consequently (D14) gives a Schwartz seminorm of order linear in n:

$$\boxed{|S_{n,a}(F)| \le AC^n (n!)^{p'} p_{kn}(F), \quad F \in \mathcal{S}_{\Delta,n}.} \tag{D15}$$

At every finite cutoff the partition decomposition agrees with the original distribution. One way to see this for lattice correlations is to note that the distribution has finite order and that the remainder

$(1-U_J)F$ tends to zero in its required test topology, using flatness. For the cutoff-uniform extension on the off-diagonal test space, the summable bound (D12) already supplies the same conclusion. Any contact distribution supported on a diagonal annihilates flat tests and cannot change (D15).

65.5 5. Consequences and limits

The usual countable-test diagonal extraction now gives subsequential limits on every flat Schwartz space, with the same growth bound. There is no requirement that S_n be represented by a function. Any finite list of species with a common local bound can be included; countably many species can be diagonalized after their constants and normalizations have been fixed.

Reflection positivity passes for the same reason as in the earlier flat-test lemma: a finite Gram matrix converges entrywise, and the closed cone of positive semidefinite matrices contains the limit. For tensors supported in strictly positive time, their reflected products are flat across the cross-half coincidence diagonals. Internal diagonals are handled by the flatness already imposed in each factor.

This proves the distributional-growth implication needed in the corrected Osterwalder–Schrader reconstruction framework from a separated product-test estimate. Full Euclidean covariance, a clustering state, and a nonzero reflected two-point form remain separate analytic inputs.

65.6 6. The scalar application, including coarse meshes

The scalar source and chessboard construction in the marked-history chapter supplies the local input with fixed parameters

$$s = 5, \qquad b = 5, \qquad p = 9. \tag{D16}$$

Its fine-mesh argument uses a common grid with cells of side comparable to h, partitioning each source into $O(n^4)$ patches whose width is comparable to h/n. The fifth-derivative source norm costs $(n/h)^5$ per patch; the patch sums cost n^4 per source. The bound $n^{9n} \le C^n (n!)^9$ gives (D1). The independent source circles belong to occupied cells and do not require the sum of all translated sources to fit one global source ball. This part requires the mesh to satisfy $a \le ch/n$.

The missing meshes can be covered directly. The scalar plaquette insertion at spacing a, smeared in a cube of size h, has the finite-cutoff form

$$O_a(f) = \sum_p f(x_p)\, a_p(U), \qquad |a_p(U)| \le C_G. \tag{D17}$$

The sum has at most $C(1+h/a)^4$ terms, including the fixed orientation count and the local support collar. When $a > ch/n$,

$$|O_a(f) - \mathbb{E}O_a(f)| \le Cn^4 \|f\|_\infty. \tag{D18}$$

Taking a product and its expectation gives $C^n n^{4n} \prod_i \|f_i\|_\infty$, which is bounded by the right side of (D1) with (D16), since $0 < h \le 1$. Hence the scalar local estimate holds at every cutoff, not just cutoffs smaller than a previously fixed separation. In particular the dyadic near-diagonal summation in (D12) is genuinely uniform in a.

This application uses the same physical scalar insertion and centering as the source argument. It does not extend (D16) to higher-dimensional curvature operators, whose additional inverse powers of a would invalidate the elementary coarse-mesh bound (D18).

66.1 1. The precise form hypothesis

Let σ be a real, symmetric, translation-invariant functional on products of separated smooth scalar tests. Suppose its positive-time reflection form is nonnegative. Complete the polynomial quotient to a Hilbert space $\mathcal{H}$, and denote the dense polynomial classes by $[F]$. Positive time translation is e^{-tH}, spatial translation is $e^{i\mathbf{x}\cdot\mathbf{P}}$, and their joint spectral measure obeys

$$E \geq 0, \qquad |q_j| \leq E \quad (j = 1, 2, 3). \tag{SP1}$$

The cube cone in (SP1) is the consequence of coordinate and diagonal lattice mirrors proved separately. Coordinate reflection positivity alone would not supply it.

Fix $0 < c < 1/64$, a finite integer s, and $p \geq 0$. The required insertion hypothesis is the following bound, uniformly in the number of factors and coefficients of the surrounding polynomials:

$$|\sigma(\theta\overline{F} \cdot (h) \cdot G)| \leq C\eta^{-p}\|h\|_{C^s}\|[\tau_{-\eta e_4/4}F]\|\|[\tau_{-\eta e_4/4}G]\|. \tag{SP2}$$

Here $0 < \eta \leq 1$, the entries of F, G lie strictly above time $\eta/4$, and h is supported in a ball of radius $c\eta$ whose center has time coordinate of absolute value at most $\eta/8$. The estimate initially for real h extends to complex h with a factor two. The constant is independent of the outside polynomials. Separate bounds for each correlation would not imply (SP2).

For the scalar action source, the strictly local fourth-jet contact subtraction gives a C^5 source ball, so the application below has $s = 5$. Only this finite derivative order matters; a C^1 source theorem is unnecessary.

66.2 2. Bounded forms and complex translation

For a test h supported near the origin as in (SP2), set

$$b_{h,\eta}(A, B) = \sigma(\theta\overline{\tau_{\eta e_4/4}A} \cdot (h) \cdot \tau_{\eta e_4/4}B).$$

The bound (SP2) implies that this form vanishes on either null space and extends continuously to $\mathcal{H} \times \mathcal{H}$. The Riesz representation theorem supplies a bounded operator $B_{h,\eta}$, with

$$\|B_{h,\eta}\| \le C\eta^{-p}\|h\|_{C^s}. \tag{SP3}$$

Its action on the dense polynomial classes is identified by testing against every such class and using a common real time translation:

$$B_{h,\eta}[A] = [(\tau_{\eta e_4/4}h)\cdot \tau_{\eta e_4/2}A]. \tag{SP4}$$

Every product in (SP4) has a strictly positive support gap. Thus the formula proves the action of an already bounded operator; it does not assume a point field on the quotient.

For $z=(z_4,\mathbf{z})\in\mathbb{C}^4$, define

$$T(z)=\int e^{-z_4E+i\mathbf{z}\cdot\mathbf{q}}\,d\Pi(E,\mathbf{q}). \tag{SP5}$$

It has norm at most one when $\operatorname{Re} z_4 \ge \sum_j |\operatorname{Im} z_j|$. In the strict interior it is holomorphic in operator norm. Indeed, on a neighborhood with margin ϵ, an order-k spectral derivative is bounded by $E^k e^{-\epsilon E}\le k!\epsilon^{-k}$. Its Taylor series therefore converges uniformly under the spectral integral on a smaller neighborhood.

For h supported in the ball of radius $c\eta/2$, put

$$M_{h,\eta}(z)=T(\eta e_4/8+z)B_{h,\eta/2}T(\eta e_4/8-z). \tag{SP6}$$

This is operator-norm holomorphic on

$$|\operatorname{Re} z_4|+\sum_j |\operatorname{Im} z_j|<\eta/8, \tag{SP7}$$

with norm at most $C\eta^{-p}\|h\|_{C^s}$. There is no restriction on real spatial translations. For real x, testing (SP6) against the downshifted classes of F,G recovers exactly the insertion $\sigma(\theta\overline{F}\cdot(\tau_x h)\cdot G)$. This follows by substituting (SP4), moving the two real translation factors to the outside polynomials, and applying one common time translation. The time factors are respectively $e^{-(\eta/8+x_4)H}$ and $e^{-(\eta/8-x_4)H}$, so the positive margins are explicit.

66.3 3. A compact parametrix adapted to the fifth-derivative norm

Use the distributional Laplacian $\Delta=\sum_{i=1}^4\partial_i^2$. In four dimensions

$$G_5(x)=-\frac{|x|^6\log|x|}{73728\pi^2},\qquad G_5(0)=0,\qquad \Delta^5 G_5=\delta_0. \tag{SP8}$$

Here $G_5\in C^5$. To verify both the coefficient and the distribution identity, use

$$\Delta(r^\alpha\log r)=\alpha(\alpha+2)r^{\alpha-2}\log r+(2\alpha+2)r^{\alpha-2}$$

successively. The first four Laplacians of $r^6\log r$ have last singular term $18432r^{-2}$. Those first four differentiations have vanishing small-sphere boundary terms, because all earlier singularities are locally

integrable with the corresponding vanishing fluxes. Finally $\Delta r^{-2} = -4\pi^2\delta_0$, by the flux through a small three-sphere. This proves (SP8).

Choose an even radial cutoff χ, supported in the unit ball and equal to one near zero. Set

$$n = \chi G_5, \qquad s_0 = \delta_0 - \Delta^5 n.$$

All derivatives of χ occur away from zero, so s_0 is a smooth compactly supported function. For $0 < r \le 1$, define

$$n_r(x) = r^6 n(x/r), \qquad s_r(x) = r^{-4} s_0(x/r).$$

Then, as compactly supported distributions,

$$\Delta^5 n_r + s_r = \delta_0, \qquad \|n_r\|_{C^5} \le Cr, \qquad \|s_r\|_{C^5} \le Cr^{-9}. \tag{SP9}$$

The first estimate uses $r^{6-k} \le r$ for $k \le 5$. No logarithmic factor in r appears: the scaled compact parametrix is defined from the fixed function n, rather than by multiplying the unscaled logarithmic expression by a new cutoff.

Take $r = c\eta/8$. On a smaller complex neighborhood of the real time slab define

$$\mathcal{O}_\eta(z) = \Delta_z^5 M_{n_r,\eta}(z) + M_{s_r,\eta}(z). \tag{SP10}$$

The compact C^5 function n_r is admitted by continuity: smooth mollification converges in its full C^5 norm and stays inside a slightly enlarged support, still strictly within the allowed ball. Equation (SP3) therefore gives an operator-norm limit, and the same bound on smaller complex neighborhoods preserves holomorphy. Thus (SP10) does not presume that the original test algebra already contained nonsmooth kernels.

Cauchy's estimate for the ten derivatives in Δ^5, together with (SP3), (SP7), and (SP9), gives

$$\boxed{\|\mathcal{O}_\eta(z)\| \le C\eta^{-(p+9)}.} \tag{SP11}$$

The constants in the finite sum defining Δ^5 are fixed. The first term costs $\eta^{-10}\eta^{-p}r$; the second costs $\eta^{-p}r^{-9}$. Both have exactly the power displayed.

66.4 4. Why this is the insertion distribution

Let f have support in a sufficiently small real ball in the slab, so that all its translates of n_r, s_r remain in the admitted insertion region. Apply the compact distribution identity (SP9), convolve with f, and use the continuity in the C^5 test norm. Fubini is legitimate for the compact supports and bounded operator-valued functions in (SP6). Integration by parts moves Δ^5 from f to $M_{n_r,\eta}$. Thus

$$\mathcal{I}_\eta(f) = \int f(x)\mathcal{O}_\eta(x)\,dx, \qquad \mathcal{I}_\eta(f) := M_{f,\eta}(0), \tag{SP12}$$

as a weak operator identity between the same time-shifted states. Equivalently, testing against any pair of polynomial classes gives the original scalar insertion distribution. Their density then identifies the bounded operator on the whole Hilbert space.

Equation (SP12) is independent of the chosen compact parametrix: two choices represent the same distribution on all smooth local tests, and their analytic densities agree there, hence on their common connected analytic domain. A finite partition extends the identity to every compact test inside a slab with a fixed positive time margin. There is no limit of unsandwiched point operators.

In particular,

$$\|\mathcal{I}_\eta(f)\| \le C\eta^{-(p+9)}\|f\|_{L^1}. \tag{SP13}$$

Smearing two transverse coordinates by a fixed φ gives an analytic planar insertion density with bound $C\eta^{-(p+9)}\|\varphi\|_{L^1}$. The constant is still independent of every surrounding field polynomial. This is the operator-strength plate bound needed by the scalar angular argument.

67.2 2. The axis kernel and its positive spectral measure

The point-kernel consequence of (A2)–(A3) can be obtained from the companion C^5 scalar insertion density. A short independent two-dimensional construction is useful for specifying exactly what is needed.

Fix a real $\varphi \in C_c^5(\mathbb{R}_z^2)$. The map

$$V_\varphi(\alpha) = [\alpha \otimes \varphi] \tag{A7}$$

is a Hilbert-valued distribution of order at most five on the positive y^0 half-plane. Indeed (A2), applied to a fixed positive-time compact set and its reflected set, bounds its norm by a constant times $\|\alpha\|_{C^5}$.

In two dimensions

$$g_4(y) = \frac{|y|^6 \log|y|}{4608\pi}, \qquad g_4(0) = 0, \qquad \Delta^4 g_4 = \delta_0. \tag{A8}$$

It belongs to C^5. Direct differentiation gives $\Delta^3(|y|^6 \log|y|) = 2304 \log|y| + 4224$, and $\Delta \log|y| = 2\pi\delta_0$. The preceding differentiations have vanishing small-circle boundary terms. Choose a smooth radial cutoff χ, equal to one near zero and supported in the unit disk. Then $n = \chi g_4$ and $s = \delta_0 - \Delta^4 n$ are respectively a compact C^5 function and a smooth compact function. The rescalings

$$n_r(y) = r^6 n(y/r), \qquad s_r(y) = r^{-2} s(y/r) \quad \text{satisfy} \quad \Delta^4 n_r + s_r = \delta_0. \tag{A9}$$

For $a > 0$ and $r < a/8$, define near $w = 0 \in \mathbb{C}^2$

$$\begin{aligned} Z_{\varphi,a}(w) = {} & \Delta_w^4 T(a/2 + w^0, w^1) V_\varphi(\tau_{ae_0/2} n_r) \\ & + T(a/2 + w^0, w^1) V_\varphi(\tau_{ae_0/2} s_r). \end{aligned} \tag{A10}$$

The cone gives a bounded holomorphic spectral integral for $T(t,x)$ when $\operatorname{Re} t > |\operatorname{Im} x|$. Every derivative on a smaller tube is bounded by $C_k E^k e^{-\epsilon E}$ under the spectral integral. Thus (A10) is a Hilbert-valued holomorphic function. For real w, the translation rule and (A9) show, after testing against a small planar test α, that

$$\int \alpha(w) Z_{\varphi,a}(w)\, dw = V_\varphi(\tau_{ae_0} \alpha). \tag{A11}$$

The identity follows by compact convolution and integration by parts. It also proves independence of r, χ: different choices represent the same distribution and hence the same continuous density.

These local densities patch to point vectors $\zeta_{\varphi,a}$, with

$$\zeta_{\varphi,b} = e^{-(b-a)H} \zeta_{\varphi,a} \quad (b > a). \tag{A12}$$

Testing their scalar products against compact planar tests recovers the original reflection form. Consequently there are continuous functions $G_{\varphi,\psi}$ on $\mathbb{R}^2 \setminus \{0\}$ such that, for planar tests with disjoint supports,

$$\begin{aligned} S_2(\alpha \otimes \varphi, \beta \otimes \psi) &= \iint \alpha(y) \beta(y') G_{\varphi,\psi}(y' - y)\, dy\, dy', \\ G_{\varphi,\psi}(t,x) &= \langle \zeta_{\varphi,a}, e^{-(t-2a)H + ixP} \zeta_{\psi,a} \rangle \quad (t > 2a). \end{aligned} \tag{A13}$$

The positive-time kernel extends to all nonzero planar points using quarter-turn symmetry. Local approximate identities show that these extensions agree on overlaps and are unique. This construction does not assume an unsandwiched point field.

For equal real labels write $G_\varphi = G_{\varphi,\varphi}$. Its kernel inherits

$$G_\varphi(R_{\pi/2} y) = G_\varphi(y), \qquad G_\varphi(t,-x) = G_\varphi(t,x), \qquad G_\varphi(y) \in \mathbb{R}. \tag{A14}$$

Let Π be the joint spectral measure in (A3). The positive measure

$$d\mu_\varphi(E,q) = e^{2aE}\, d\langle \zeta_{\varphi,a}, \Pi(E,q) \zeta_{\varphi,a} \rangle \tag{A15}$$

is independent of $a > 0$, by (A12). It is supported on $\mathcal{C} = \{E \geq |q|\}$, is locally finite, and has finite Laplace weights at every positive time:

$$\begin{aligned} G_\varphi(t,x) &= \int e^{-tE+ixq}\, d\mu_\varphi, \\ \Gamma_\varphi(t) &= \int e^{-tE}\, d\mu_\varphi = G_\varphi(t,0) < \infty, \qquad t > 0. \end{aligned} \tag{A16}$$

In particular $\Gamma_\varphi \geq 0$ is nonincreasing. After a quarter turn, every nonzero y has positive time coordinate at least $|y|/\sqrt{2}$. Hence

$$|G_\varphi(y)| \leq \Gamma_\varphi(|y|/\sqrt{2}). \tag{A17}$$

For every positive-time planar test α, (A13)–(A16) give the exact plate norm

$$\|[\alpha \otimes \varphi]\|^2 = \int \left| \iint \alpha(t,x) e^{-tE+ixq}\, dt\, dx \right|^2 d\mu_\varphi. \tag{A18}$$

If $t \geq \delta > 0$ on the test support, the absolute integral before interchanging variables is at most $\|\alpha\|_1^2 \Gamma_\varphi(2\delta)$. This proves the required Fubini step even when μ_φ has infinite total mass.

67.4 4. A dilation average with no Fourier-zero assumption

For $h>0$, $1\le\lambda\le 2$, set

$$\alpha_h^\lambda(t,x)=a(t/h)b(4x/(\lambda h)),\qquad f_h^\lambda=\alpha_h^\lambda\otimes\varphi. \tag{A30}$$

Its transform in (A18) is the product of

$$A_h(E) = \int a(t/h)e^{-tE}dt \geq \tfrac{1}{2}he^{-3hE},$$
$$B_{h,\lambda}(q) = \frac{c_b\lambda h}{4}\left(\frac{\sin(\lambda hq/32)}{\lambda hq/32}\right)^8. \tag{A31}$$

There is an explicit $c_B > 0$ such that

$$\int_1^2 |B_{h,\lambda}(q)|^2 d\lambda \geq c_B h^2(1+h^2q^2)^{-8}. \tag{A32}$$

To check every Fourier zero, write $x = h|q|/32$. For $0 \leq x \leq 1$, the sinc is at least $m_0 = \sin 2/2 > 0$, so the left side is at least $c_b^2h^2m_0^{16}/16$. For $x \geq 1$, it is at least

$$\frac{c_b^2h^2}{2^{18}}x^{-17}\int_x^{2x}\sin^{16} v\, dv. \tag{A33}$$

Let

$$m_{16} = \min_{s\in[0,\pi]}\int_s^{s+1}\sin^{16} v\, dv > 0. \tag{A34}$$

The minimum is positive because the integrand cannot vanish on an interval. An interval of length $x \geq 1$ contains at least $x/2$ disjoint unit intervals, so (A33) is at least $c_b^2 2^{-19}m_{16}h^2x^{-16}$. One may therefore take

$$c_B = \min\{c_b^2m_0^{16}/16,\, c_b^2m_{16}/2^{19}\}. \tag{A35}$$

Both regimes imply (A32), since $h^2q^2 = 1024x^2$.

Define the finite numerical constant

$$C_8 = \sup_{r\geq 0} e^{-r}(1+r^2)^8, \qquad c_{\mathrm{pl}} = c_B/(4C_8) > 0. \tag{A36}$$

Because $|q| \leq E$,

$$e^{-6hE}(1+h^2q^2)^{-8} \geq C_8^{-1}e^{-7hE}. \tag{A37}$$

The norm identity (A18), nonnegative Fubini, and (A31)–(A37) now give

$$\int_1^2 \|[f_h^\lambda]\|^2 d\lambda \geq c_{\mathrm{pl}}h^4\Gamma_\varphi(7h). \tag{A38}$$

In particular, increasing the spline regularity has not increased the power of h in this estimate. The extra polynomial Fourier decay was paid by one additional fixed exponential weight using the spectral cone.

67.5 5. Positive pedestals and the critical ultraviolet growth

Choose $R \geq 1$ with $\operatorname{supp}\varphi \subset B(0,R)$. For $0 < h \leq \min(1, 2d_0/5)$, put

$$A_h = \sum_{|\gamma|\leq 5} h^{|\gamma|}\|\partial^\gamma\varphi\|_\infty. \tag{A39}$$

For a nonzero label this is positive, and for $h \leq 1$ it is bounded by the C^5 norm defined as the sum of these sup norms. Use the finite set $\mathcal{I}_h = \{i : \psi(z/h - i)\varphi(z) \not\equiv 0\}$. It has cardinality at most

$$N_h \leq (2R/h+3)^2 \leq 25R^2h^{-2}. \tag{A40}$$

For $i \in \mathcal{I}_h$, define

$$\begin{aligned} \varphi_i^- &= 2A_h\widetilde{\psi}(z/h - i), \\ \varphi_i^+ &= 2A_h\widetilde{\psi}(z/h - i) + \psi(z/h - i)\varphi(z), \\ h_i^\pm &= \alpha_h^\lambda \otimes \varphi_i^\pm. \end{aligned} \tag{A41}$$

Then

$$f_h^\lambda = \sum_{i\in\mathcal{I}_h} (h_i^+ - h_i^-), \qquad A_h\widetilde{\psi}(z/h - i) \le \varphi_i^\pm \le 3A_h\widetilde{\psi}(z/h - i). \tag{A42}$$

The tests $h_i^\pm$ are nonnegative C_c^5 functions, with $A_h \le \|h_i^\pm\|_\infty \le 3A_h$.

Here are all shape checks. Set $d = 5h/2$. Each support is contained in the radius-d ball centered at $(5h/2, 0, hi)$: bounds $h/2, h/2, 3h/2, 3h/2$ on the coordinate half-widths give squared radius at most $5h^2 < d^2$. Its reflected partner has the same shape. Their support distance lies between $4h$ and $6h$, hence between d and $6d$. The exact integrals of the three fixed factors give

$$\int h_i^\pm \ge \frac{125c_b}{128}A_h h^4 \ge \frac{c_b}{120}\|h_i^\pm\|_\infty d^4. \tag{A43}$$

For a derivative of total order $1 \le k \le 5$, the product rule and (A39) give

$$h^k\|\partial^\gamma h_i^\pm\|_\infty \le A_h C_{\mathrm{sh}}. \tag{A44}$$

The definition (A27) therefore supplies the derivative condition in (A6), with a fixed factor of spare margin. All constructed reflected pairs belong to the exact ultraviolet class, uniformly in h, λ, the label φ, and the cell index.

Reflection positivity, the Hilbert triangle inequality, and (A5) yield

$$\|[f_h^\lambda]\| \le 6N_h A_h\sqrt{u(5h/2)}. \tag{A45}$$

Squaring, integrating $1 \le \lambda \le 2$, and using (A38), (A40) proves the explicit plate estimate

$$\Gamma_\varphi(7h) \le \frac{22500}{c_{\mathrm{pl}}}\, R^4\|\varphi\|_{C^5}^2 h^{-8}u(5h/2). \tag{A46}$$

Thus

$$t^8\Gamma_\varphi(t) \longrightarrow 0 \quad (t \downarrow 0). \tag{A47}$$

The zero transverse label is immediate. The power eight in (A46) comes from the h^4 transform normalization in (A38) and the square of the $O(h^{-2})$ transverse cell count. It does not come from a point-field bound or from differentiating the ultraviolet hypothesis.

67.6 6. Entire angular continuation and the three possible harmonics

Fix $r > 0$ and a real label φ. On $|\operatorname{Re} z| < \pi/2$, define

$$g_r^{(0)}(z) = \int e^{-rE\cos z + irq\sin z}\, d\mu_\varphi(E, q). \tag{A48}$$

If $z = v + is$, the absolute multiplier is

$$\exp[-r\cos v(E\cosh s + q\sinh s)] \le e^{-r\cos v\, e^{-|s|}E}. \tag{A49}$$

This proves holomorphy by exponential domination on each compact substrip. On real points it is $G_\varphi(r\cos v, r\sin v)$.

Translate this formula by each integer multiple of $\pi/2$. Quarter-turn symmetry (A14) makes the formulas agree on real overlaps. The identity theorem extends that agreement to their connected complex overlaps, so they patch to an entire, $\pi/2$-periodic function g_r. For every real v, s, choose a quarter turn that puts the real angle within $\pi/4$ of zero. Then

$$|g_r(v+is)| \le \Gamma_\varphi(re^{-|s|}/\sqrt{2}) = o(e^{8|s|}), \tag{A50}$$

uniformly in v, by (A47).

For the Fourier coefficients

$$a_m(r) = \frac{2}{\pi}\int_0^{\pi/2} g_r(v)e^{-4imv}\,dv, \tag{A51}$$

shift the integration contour to height s. The vertical sides cancel by periodicity, giving

$$|a_m(r)| \le e^{4ms}\sup_v |g_r(v+is)|. \tag{A52}$$

For $m \ge 2$ take $s \to -\infty$; for $m \le -2$, take $s \to +\infty$. The little-o in (A50) kills the endpoint modes $|m| = 2$ as well as the higher ones. Fourier uniqueness for continuous periodic functions, for example by Fejer approximation, gives

$$g_r(z) = a_0(r) + a_1(r)e^{4iz} + a_{-1}(r)e^{-4iz}. \tag{A53}$$

Reality and the tangential reflection imply these coefficients are real and $a_1 = a_{-1}$, although the final argument does not require that equality. An $O(t^{-8})$ premise would not suffice to remove the ± 8 angular modes; the vanishing ultraviolet strength is used at precisely this point.

67.7 7. Constructing the diagonal positive measure without assuming its cone

Set

$$D(t,x) = G_\varphi(R_{\pi/4}(t,x)), \qquad t > 0. \tag{A54}$$

Introduce formal point vectors $e_{(t,x)}$, with

$$\langle e_{(t,x)}, e_{(t',x')}\rangle_D = D(t+t', x'-x). \tag{A55}$$

This kernel is positive semidefinite. To prove it, replace each point $R_{\pi/4}(t,x)$ by a small radial smooth planar mollifier, tensor with φ, and apply (A4) to an arbitrary complex linear combination. For sufficiently small mollifiers every support remains strictly on the positive side of the diagonal mirror. Continuity of the kernel in (A13) then gives (A55) by taking the mollifier radii to zero. No uniform bound on the C^5 norms of the approximate identities is needed for this finite-dimensional positivity limit.

On finite sums of point vectors, define

$$T_s^D e_{(t,x)} = e_{(t+s,x)}, \qquad U_b^D e_{(t,x)} = e_{(t,x+b)}. \tag{A56}$$

The second family preserves the form and has inverses. The first is symmetric and has the semigroup law. For $v = \sum_i c_i e_{(t_i,x_i)}$, put $\delta = 2\min_i t_i > 0$. Equation (A17) bounds its entire time orbit:

$$\|T_s^D v\|_D^2 \le \Gamma_\varphi(\delta/\sqrt{2})\left(\sum_i |c_i|\right)^2 \quad (s \ge 0). \tag{A57}$$

Moreover

$$\|T_s^D v\|_D^2 = \langle v, T_{2s}^D v\rangle_D \le \|v\|_D \|T_{2s}^D v\|_D. \tag{A58}$$

This first preserves the null space, then proves contraction by iterating to $2^n s$ and using the bounded orbit (A57). Thus T_s^D extends to a self-adjoint contraction on the completed quotient. It is positive since $T_s^D = (T_{s/2}^D)^2$. Continuity of D gives strong continuity on point vectors and then on the completion. The two families commute.

The resulting strongly commuting generators are $T_s^D = e^{-sH_D}$, $U_b^D = e^{ibP_D}$, with $H_D \ge 0$. For an elementary construction, T_1^D is injective: its kernel is the kernel of every positive dyadic root, and strong continuity then forces such a vector to vanish. Define $H_D = -\log T_1^D$ by spectral calculus; the positive-root identity and continuity recover every time s. The unitary group gives P_D, commuting with that spectral calculus.

If Π_D is their joint spectral measure, the formula

$$d\nu(E,q) = e^{2aE}\, d\langle e_{(a,0)}, \Pi_D(E,q) e_{(a,0)}\rangle_D \tag{A59}$$

defines a positive measure independent of $a > 0$. It is carried by $[0,\infty)\times\mathbb{R}$, and

$$D(t,x) = \int e^{-tE+ixq}\, d\nu, \qquad \int e^{-tE}\, d\nu = D(t,0) < \infty. \tag{A60}$$

There is no diagonal spectral-cone assumption in this construction. That cone will be a consequence of the next positive-measure identity.

67.8 8. A positive-measure identity removes the remaining modes

We use a simple uniqueness statement. If two positive measures on $[0,\infty)\times\mathbb{R}$ have finite Laplace weights at every positive time and identical transforms $\int e^{-tE+ixq} d\rho$ for $t > 0, x \in \mathbb{R}$, they coincide. For each real x, the transforms extend holomorphically to $\operatorname{Re} t > 0$. Equality there, at $t = a + i\tau$, identifies the joint Fourier transforms of the finite measures $e^{-aE}\rho$. Finite Fourier transforms determine their measures: convolve their difference with a Gaussian, apply its Fourier integral and Fubini, then take the approximate-identity limit on continuous compact tests. This proof also applies to finite positive linear combinations of such measures.

For $s \in \mathbb{R}$, let

$$L_s(E,q) = (E\cosh s + q\sinh s,\ E\sinh s + q\cosh s), \qquad \mu_s = (L_s)_*\mu_\varphi. \tag{A61}$$

These are positive measures supported on $\mathcal{C}$, with finite Laplace weights, since the new energy is at least $e^{-|s|}E$. Define

$$F_s(t,x) = \int e^{-tE+ixq}\, d\mu_s. \tag{A62}$$

The substitution in (A61) gives, for $(t,x) = R_\theta(r,0)$, $t > 0$,

$$F_s(t,x) = g_r(\theta + is). \tag{A63}$$

One may check this without an angular continuation assumption for the diagonal measure: $R_{is}(t,x)$ is in the axis tube because its real time is $t\cosh s$ and the magnitude of its imaginary space coordinate is $t|\sinh s|$. Its spectral multiplier is exactly the one in (A62).

Put $C = \cosh 4 > 1$. Each of the three terms in (A53) satisfies

$$(C-1)g_r(\theta + \pi/4) + g_r(\theta + i) + g_r(\theta - i) = (C+1)g_r(\theta). \tag{A64}$$

The constant term has coefficient $C+1$ on each side. A term with frequency 4 or -4 changes sign under $\pi/4$, and its two imaginary translates sum to $2C$ times that term. This verifies the identity directly.

Apply the transform uniqueness just proved to (A64), using (A16), (A60), and (A63). It gives an equality of POSITIVE measures:

$$(C-1)\nu + \mu_1 + \mu_{-1} = (C+1)\mu_\varphi. \tag{A65}$$

Every coefficient in this equality is positive. Therefore

$$0 \le \nu \le \frac{C+1}{C-1}\mu_\varphi, \tag{A66}$$

and ν is supported on the exact cone $\mathcal{C}$. This deduction is where diagonal reflection positivity enters decisively; a diagonal cone was not presumed.

For arbitrary $s \in \mathbb{R}$, the function

$$w_s(E,q) = e^{-r(E\cosh s + q\sinh s)} \tag{A67}$$

is positive and integrable against every measure in (A65). For ν, use (A66); for $\mu_{\pm1}$, use $L_sL_{\pm1} = L_{s\pm1}$. Integrating (A65) against w_s, and substituting (A53), gives

$$\begin{aligned} F_s(r,0) &= a_0 + a_1e^{-4s} + a_{-1}e^{4s} \ge 0,\\ \int w_s\,d\nu &= a_0 - a_1e^{-4s} - a_{-1}e^{4s} \ge 0. \end{aligned} \tag{A68}$$

Divide both inequalities by e^{4s} and take $s \to +\infty$. They give $a_{-1} \ge 0$ and $a_{-1} \le 0$. Taking $s \to -\infty$ gives both signs for a_1. Thus $a_1 = a_{-1} = 0$, and G_φ is radial.

67.9 9. Mixed labels, full two-point covariance, and limitations

Symmetry and the planar half-turn give

$$G_{\varphi,\psi}(y) = G_{\psi,\varphi}(-y) = G_{\psi,\varphi}(y). \tag{A69}$$

For real labels, polarization therefore yields

$$G_{\varphi,\psi} = \tfrac{1}{4}(G_{\varphi+\psi} - G_{\varphi-\psi}). \tag{A70}$$

Both terms are radial, and complex labels follow by bilinear extension. This proves the theorem, including every mixed angular coefficient.

If the companion insertion-density theorem has supplied the continuous four-dimensional density $C(x)$ for $x \ne 0$, the preceding result implies its invariance under rotations in this coordinate plane. Indeed for $y \ne 0$,

$$G_{\varphi,\psi}(y) = \iint \varphi(z)\psi(z')C(y, z'-z)\,dz\,dz'. \tag{A71}$$

Approximating transverse point masses by compact C^5 tests and using continuity shows $C(R_\theta y, z) = C(y,z)$. If $y = 0$ and the full displacement is nonzero, the same assertion follows by continuity from $y \ne 0$. Hypercubic conjugation applies the argument to every coordinate plane. Their rotations generate $SO(4)$; together with the already assumed reflections, C is $O(4)$ invariant off the diagonal.

Testing this continuous density gives the same assertion for arbitrary disjoint four-dimensional supports. No possible distribution supported on an exceptional planar projection is silently discarded in this passage: it uses the actual full off-diagonal density as stated.

The application to the Wilson limit is conditional on the physical inputs (A2)–(A6). The C^5 source theorem and compact insertion parametrix supply the regularity/kernel part; the coordinate and diagonal physical reflections supply the positive forms; the actual spectral cube cone supplies (A3); and the finite-window physical comparison supplies the uniform vanishing function u on the specified shape class. The proof here is independent of how those inputs are obtained. It proves no large-field preparation theorem, no all-orders angular closure, and no mass gap.

68.1 1. Inputs and the precise scalar test algebra

Let S be a centered real symmetric scalar moment functional on finite tuples of pairwise separated compact C^5 tests in $\mathbb{R}^4$, and on their finite complex linear combinations. The empty tuple has value one. Assume:

1. Translation and hypercubic invariance, local C^5 continuity for each fixed tuple length, and the off-diagonal distributions supplied by the all-cutoff flat-test bound.
2. Reflection positivity in every coordinate Hilbert picture, with its translation dictionary, and commuting generators satisfying $H_j \geq 0$, $|P_{j,r}| \leq H_j$ for every $r \neq j$.
3. The operator-strength scalar plate bound obtained from the C^5 source insertion and its compact Δ^5 parametrix. Its time-margin exponent is a fixed q; the present construction supplies $q = 9$.
4. The separate two-point angular elimination: for real smooth transverse labels, the two-point angular coefficient of every nonzero frequency vanishes. That result uses the actual ultraviolet shape law and the diagonal reflection picture.

The first three inputs produce the analytic and finite-band objects in Sections 2–5. The fourth is used only in Section 7. All Hilbert polynomials in the continuum argument belong to the admitted separated algebra and then its proved flat completion. Unrenormalized products of overlapping same-side tests are not assumed to exist.

Fix the plane $y = (x_4, x_1)$, write $z = (x_2, x_3)$, and let $\theta(y^0, y^1) = (-y^0, y^1)$. A label φ is a real compact smooth function of z. In the x_4 Hilbert picture write $\Omega = [1]$, $H = H_4$, $P = P_{4,1}$. Then

$$e^{-tH}[F] = [\tau_{te_4}F], \qquad e^{ixP}[F] = [\tau_{xe_1}F], \qquad H \geq |P|, \qquad H\Omega = P\Omega = 0. \tag{SA1}$$

68.2 2. Bounded point plates and actual ordered product vectors

The analytic insertion density gives bounded operators $B_{\varphi,a}$, $0 < a \leq 1/4$, with

$$\|B_{\varphi,a}\| \leq Ca^{-q}\|\varphi\|_1, \qquad B_{\varphi,b} = e^{-(b-a)H}B_{\varphi,a}e^{-(b-a)H} \quad (b > a). \tag{SA2}$$

These are time-regularized plane-point insertions. Concretely they are the operator-norm limit of the bounded insertions of $\alpha_\epsilon(y)\varphi(z)$, with α_ϵ a unit-mass smooth time-even plane approximate identity. The analytic density is norm continuous in y, so that limit is valid even though the C^5 norm of α_ϵ diverges. Its L^1 mass stays one. The limit therefore uses the proved plate bound, not a uniform estimate on those diverging source norms.

For real φ, reflection of the time-even approximants gives $B^*_{\varphi,a} = B_{\varphi,a}$. The padding identity in (SA2) follows first for smooth profiles by testing on the dense polynomial classes and making one common real time translation. Pass to the operator-norm limit to obtain it for point plates. These observations give the exact meaning of the mnemonic $B_{\varphi,a} = e^{-aH}\phi_\varphi(0)e^{-aH}$; no unsandwiched point field is postulated.

For $v \in \mathbb{C}^2$ put

$$v_\pm = v^0 \pm iv^1, \quad \mathbb{V} = \{\operatorname{Re} v_+ > 0,\ \operatorname{Re} v_- > 0\}, \quad v^* = (\overline{v^0}, -\overline{v^1}). \tag{SA3}$$

The spectral theorem and $H \geq |P|$ define

$$T(v) = e^{-v^0 H + iv^1 P}, \qquad \|T(v)\| \leq 1, \quad T(v)^* = T(v^*), \quad T(v)T(w) = T(v+w). \tag{SA4}$$

Here v, w lie in the closed tube when necessary. The map is operator-norm holomorphic in $\mathbb{V}$: on a neighborhood with light-coordinate real parts at least $r > 0$, its order-m spectral derivatives are bounded by $C_m E^m e^{-rE} \leq C'_m r^{-m}$. The exponential power series then converges uniformly under the spectral integral on a smaller neighborhood.

Let

$$\mathbb{D}^+_n = \{(y_1, \dots, y_n) : y_1 \in \mathbb{V},\ y_{i+1} - y_i \in \mathbb{V}\}.$$

For a sufficiently small a depending on these strict tube margins define the ACTUAL Hilbert vector

$$\begin{aligned}\Psi_a(\mathbf{y}; \vec{\varphi}) = T(y_1 - ae_0)B_{\varphi_1,a}T(y_2 - y_1 - 2ae_0)B_{\varphi_2,a}\cdots \\ T(y_n - y_{n-1} - 2ae_0)B_{\varphi_n,a}\Omega.\end{aligned} \tag{SA5}$$

Padding (SA2) and the semigroup rule show that this is independent of every admissible choice of a. Thus it defines a Hilbert-valued holomorphic vector Ψ on $\mathbb{D}^+_n$. Set $\Psi(\varnothing) = \Omega$. For real time-ordered points it is the strong limit of the corresponding smoothed separated product vectors: insert the smooth approximants in (SA5), identify the classes by the translation dictionary, and use operator norm convergence of the finitely many bounded factors.

Similarly, for $v_i \in \mathbb{V}$ define

$$K_n(v; \vec{\varphi}) = \langle \Omega, B_{\varphi_1,a}T(v_1 - 2ae_0)B_{\varphi_2,a}\cdots T(v_{n-1} - 2ae_0)B_{\varphi_n,a}\Omega\rangle. \tag{SA6}$$

It is independent of a, holomorphic on $\mathbb{V}^{n-1}$, and is the density of the original moment on strictly time-ordered plane profiles:

$$S(\alpha_i \otimes \varphi_i : 1 \leq i \leq n) = \int \prod_i \alpha_i(y_i)\, K_n(y_2 - y_1, \dots, y_n - y_{n-1}; \vec{\varphi})\, d\mathbf{y}. \tag{SA7}$$

One proves (SA7) first for small smooth bumps using the vector limit just described, then by Riemann integration and local C^5 continuity for arbitrary compact profiles with a fixed time gap. All integrals have compact support strictly inside one ordered domain. Complex labels follow by multilinearity.

Explicitly, use the same plane approximate identity β_ϵ in the entries and write $\alpha_i * \beta_\epsilon = \int \alpha_i(y)\tau_y\beta_\epsilon\, dy$. For each fixed ϵ, Riemann sums converge in C^5 on one compact support, so multilinearity and

the separated continuity give the integral identity with the smoothed bounded-operator kernel. Its convergence to (SA6) is locally uniform on the chosen ordered compact set, by operator-norm convergence of its finitely many factors. The left side converges to the original moment because $\alpha_i * \beta_\epsilon \to \alpha_i$ in C^5. This proves (SA7) without integrating an assumed unsmeared field.

Put $d(v) = \min(1, \min_{i,\pm} \mathrm{Re}(v_i)_\pm)$. Taking $a = d(v)/4$ in (SA6) gives

$$|K_n(v;\vec{\varphi})| \le C^n d(v)^{-qn} \prod_i \|\varphi_i\|_1. \tag{SA8}$$

There is exactly one power a^{-q} per inserted field. This linear dependence of the tube exponent on n is the key quantitative input below.

68.3 3. Ordered charts, entire angle functions, and finite harmonics

Quarter-turn invariance gives the same kernels on the rotated tubes $\mathbb{V}_\nu = R_\nu \mathbb{V}$, $\nu \in (\pi/2)\mathbb{Z}$, with arbitrary point permutations and the correspondingly permuted labels. These kernels agree on every nonempty complex overlap. Indeed each ordered chart is convex and stable under taking its real part; an intersection has a real open subset. At real points in that subset, small smooth profiles have both orderings, and (SA7) represents the same original moment in both. Concentrating the profiles proves equality of the continuous densities. Holomorphic uniqueness on the connected intersection proves equality there.

They glue to a holomorphic function $\mathbf{K}_n$ on the union $\mathcal{U}_n^{\mathrm{ord}}$ of the ordered quarter-turn charts. This union is connected: a real configuration can have any prescribed strict time order and independently any prescribed strict abscissa order, so every time-order chart meets every abscissa-order chart. Its real points are the configurations for which all times are distinct OR all abscissae are distinct. The glued function is symmetric in labelled points, translation invariant, and quarter-turn invariant.

Call a real planar configuration generic if its points are distinct and no two nonzero pair differences are perpendicular. These configurations form an open dense set: the excluded relations are finitely many zero sets of nonzero real polynomials. For such a configuration, every real rotation is ordinary. Failure would give one horizontal and one vertical nonzero difference, contradicting the definition.

Define $a_{\mathbf{y}}(z)$ locally by applying $\mathbf{K}_n$ to $R_z\mathbf{y}$. This gives an entire function of z. To justify the global statement, cover the real angle axis by intervals on which a particular ordered coordinate projection is strict. Their vertical strips are admitted because

$$(R_{is}v)_\pm = e^{\mp s} v_\pm, \qquad R_{is}\mathbb{V}_\nu = \mathbb{V}_\nu. \tag{SA9}$$

On strip overlaps, the already proved real-chart identity and the one-variable identity theorem give exact gluing. Quarter-turn invariance gives period $\pi/2$.

For the fixed generic configuration define

$$\delta_{\mathbf{y}} = \min_{t\in[0,\pi/2]} \max\left\{ \min_{i<j} |(R_t(y_i - y_j))^0|, \min_{i<j} |(R_t(y_i - y_j))^1| \right\} > 0. \tag{SA10}$$

Positivity follows by compactness and the absence of simultaneous ties in both projections. Choose at each angle the projection attaining the maximum, and sort the points in that frame. Equation (SA9) and (SA8) yield

$$|a_{\mathbf{y}}(t+is)| \le A_{\mathbf{y},\vec{\varphi}} e^{qn|s|}. \tag{SA11}$$

Its Fourier coefficients have frequencies $4m$. Shifting their contour to height s, the two vertical sides cancel by periodicity, and

$$|c_m| \le A_{\mathbf{y},\vec{\varphi}} e^{4ms+qn|s|}.$$

Let $s \to -\infty$ when $4m > qn$, and $s \to +\infty$ when $4m < -qn$. Thus

$$a_{\mathbf{y}}(z) = \sum_{|m| \le \lfloor qn/4 \rfloor} c_m(\mathbf{y};\vec{\varphi}) e^{4imz}. \tag{SA12}$$

The equality follows either from the Laurent expansion in e^{4iz}, or from the Fourier series on the real axis followed by analytic uniqueness.

Here is a finite algebraic construction of these coefficients on ALL ordinary charts. Set $N = \lfloor qn/4 \rfloor$ and choose $2N+1$ distinct real numbers s_ℓ. The matrix $V_{\ell m} = e^{-4ms_\ell}$, $-N \le m \le N$, is invertible: after multiplying each row by a nonzero factor it is a Vandermonde matrix at the distinct positive numbers e^{-4s_ℓ}. Define

$$c_m(\mathbf{z}) = \sum_\ell (V^{-1})_{m\ell} \mathbf{K}_n(R_{is_\ell}\mathbf{z}). \tag{SA13}$$

Imaginary rotations preserve every ordered chart, so these are genuine holomorphic functions on $\mathcal{U}_n^{\mathrm{ord}}$. They equal the Fourier coefficients on the real generic set. Holomorphic uniqueness therefore gives

$$\mathbf{K}_n = \sum_m c_m, \qquad c_m(R_{is}\mathbf{z}) = e^{-4ms} c_m(\mathbf{z}). \tag{SA14}$$

They are symmetric and translation invariant, with $c_{-m} = \overline{c_m}$ on real configurations.

They also extend uniquely as real-analytic functions to every configuration of DISTINCT planar points, even if some coordinate projections tie. Choose an angle t for which all rotated times are distinct and define locally

$$\widetilde{c}_m(\mathbf{y}) = e^{-4imt} c_m(R_t\mathbf{y}). \tag{SA15}$$

Only finitely many angles are excluded. Two choices agree on the dense generic set of their overlap by (SA12), hence everywhere there by continuity. This proves gluing and

$$\widetilde{c}_m(R_t\mathbf{y}) = e^{4imt}\widetilde{c}_m(\mathbf{y}). \tag{SA16}$$

No density assertion for the original distribution on exceptional time ties has yet been made.

68.4 4. Boosted Gram positivity is a vector identity

For $\mathbf{x} \in \mathbb{D}_k^+$ and $\mathbf{y} \in \mathbb{D}_l^+$, let $J(\mathbf{x},\mathbf{y})$ be the consecutive differences of the reflected-left and right lists:

$$J = ((x_k - x_{k-1})^*, \dots, (x_2 - x_1)^*, x_1^* + y_1, y_2 - y_1, \dots, y_l - y_{l-1}). \tag{SA17}$$

The left labels are reversed. Directly multiply the bounded factors in (SA5), using $T(v)^* = T(v^*)$, $B_{\varphi,a}^* = B_{\varphi,a}$, and (SA6). This proves the exact equality

$$K_{k+l}(J; \text{ordered labels}) = \langle \Psi(\mathbf{x}), \Psi(\mathbf{y}) \rangle. \tag{SA18}$$

The usual omissions give the same identity for an empty list, with vector Ω. In particular it is positive for matrices containing different tuple lengths.

The crucial additional identity is

$$(R_{is}v)^* = R_{is}(v^*). \tag{SA19}$$

This uses the exact antilinear involution $v^* = -\theta\overline{v}$, not ordinary conjugation. Indeed

$$R_{is} = \begin{pmatrix} \cosh s & -i\sinh s \\ i\sinh s & \cosh s \end{pmatrix}$$

and both sides of (SA19) equal $(\cosh s\,\overline{v}^0 + i\sinh s\,\overline{v}^1,\ i\sinh s\,\overline{v}^0 - \cosh s\,\overline{v}^1)$. Together with tube invariance in (SA9), it implies

$$K_{k+l}(R_{is}J) = \langle \Psi(R_{is}\mathbf{x}), \Psi(R_{is}\mathbf{y})\rangle. \tag{SA20}$$

Thus the boosted kernels are positive semidefinite because they are Gram matrices of actual Hilbert vectors. Positivity has not been analytically continued from a positive real scalar function.

For arbitrary real point configurations X, Y in the positive half-plane, define the boosted kernel by

$$\mathsf{G}_s(X,Y) = \sum_m e^{-4ms}\widetilde{c}_m^{(|X|+|Y|)}(\theta X, Y). \tag{SA21}$$

If each individual configuration has distinct times, sort its points; (SA14) and (SA20) prove positivity. For a finite family with ties, perturb the points slightly inside the positive half-plane. The combined reflected configurations remain collision free. Real-analytic continuity of the finitely many coefficients passes positivity to the limit.

Let b_n be the largest absolute index occurring for some real labels, with $b_0 = b_1 = 0$. Then

$$b_n \le qn/4. \tag{SA22}$$

For $m_* = b_{2k}$, multiply the k-by-k configuration Gram kernel by e^{-4m_*s} and let $s \to +\infty$. Its limit

$$H_k(X,Y) = \widetilde{c}_{-m_*}^{(2k)}(\theta X, Y) \tag{SA23}$$

is positive semidefinite. Across an arbitrary oriented line with normal $R_\nu e_0$, its transported kernel is $e^{4im_*\nu}\widetilde{c}_{-m_*}(\theta_{\text{line}}X, Y)$, after the corresponding translation. The fixed unit phase is essential. This statement concerns the top coefficient, not reflection positivity of the original functional at every angle.

Gram Cauchy–Schwarz also gives

$$2b_{k+l} \le b_{2k} + b_{2l}, \qquad k, l \ge 0. \tag{SA24}$$

To prove it, take a real time-ordered configuration where the negative top coefficient of length $k+l$ is nonzero, and translate its cut between entries k and $k+1$ to the mirror. Such a configuration exists by real analyticity: a nonzero coefficient cannot vanish on that nonempty open set. The off-diagonal boosted Gram entry has leading magnitude of order $e^{4b_{k+l}s}$, while the two diagonal entries have rates at most $e^{4b_{2k}s}$ and $e^{4b_{2l}s}$. Compare exponents in Cauchy–Schwarz. For $k = 0$, use the actual empty vector Ω; its diagonal entry is one. This includes the boundary case of (SA24) needed later.

68.5 5. Polygon divisibility, including label release

We prove

$$k \mid 2b_{2k}. \tag{SA25}$$

If it failed, put $m_* = b_{2k}$ and $H = \widetilde{c}_{-m_*}^{(2k)}$. Assign the same arbitrary real label φ to the vertices of a regular $2k$-gon. Rotation through π/k permutes the labelled configuration, while (SA16) multiplies H by $e^{-4\pi i m_*/k} \ne 1$. Thus this diagonal polygon value is zero.

Call a labelled subset a zero set if every completion to $2k$ distinct points, with all free real labels arbitrary, has $H = 0$. Suppose such a subset lies in the reflected union $\theta_{\mathrm{line}}Q \cup Q$, where Q has $j \le k$ points on one side and reflected labels agree. Fill Q by $k - j$ arbitrary points A on that side. The top positive kernel (SA23), transported to this line with its unit phase, has zero diagonal at $Q \cup A$. Its entire row is therefore zero, by the positive 2×2 minor inequality. Hence

$$H(\theta_{\mathrm{line}}(Q \cup A), B) = 0 \tag{SA26}$$

for every k-point configuration B on that side.

For each fixed choice of the free labels, (SA26) holds on an open set of all free positions. Their configuration space, avoiding the fixed reflected points and all mutual collisions, is connected: it is a planar configuration space with finitely many points removed. One may move its points successively along polygonal paths avoiding the remaining finite set. Real-analytic uniqueness extends (SA26) to that whole space. Thus the reflected Q itself is a zero set, with every remaining label now unrestricted. This is the label-release step, not only a statement about equal-label diagonal values.

Apply the reduction first across an edge-bisector of the regular polygon, obtaining an arc of k vertices as a zero set. An arc of $j \le k$ vertices is contained in the reflected union of an arc of $\lceil j/2 \rceil$ vertices, with no vertex on the reflecting line. For even j, reflect indices $i \mapsto j - 1 - i$; for odd j, use $i \mapsto j - i$ and the union of $j + 1$ vertices. Both have an odd index sum, so no polygon vertex is fixed by the reflection. Repeated reduction reaches one point labelled by φ.

Translation invariance makes every point with that label a zero set. Since φ was arbitrary and the other labels have been released, H vanishes on every labelled configuration. This contradicts the definition of m_*, proving (SA25). The case $m_* = 0$ is automatic.

68.6 6. Integer convexity reduces all frequencies to two points

Set $a_k = b_{2k}$, $a_0 = 0$. Equation (SA24) gives

$$2a_k \le a_{k-1} + a_{k+1}, \qquad 0 \le a_k \le Ck. \tag{SA27}$$

The integer increments $d_k = a_k - a_{k-1}$ are nonnegative, nondecreasing, and bounded. They therefore eventually equal one integer d. For large k,

$$a_k = dk - D, \qquad D = \sum_{j \ge 1} (d - d_j) \ge 0, \tag{SA28}$$

with finitely many nonzero summands. Divisibility (SA25) says that every sufficiently large k divides $2D$. Hence $D = 0$, all increments equal $d = a_1$, and

$$b_{2k} = kb_2, \qquad 2b_n \le b_{2n} = nb_2. \tag{SA29}$$

The separate two-point elimination now gives $b_2 = 0$, so every $b_n = 0$. Every ordinary-chart scalar kernel is rotation invariant in the chosen plane. No positivity or regularity for higher composite gauge fields has entered this scalar argument.

68.7 7. Removing possible distributions on exceptional projections

Let T_n be the actual off-diagonal distribution on $\mathcal{C}_n = \{(x_1, \dots, x_n) : x_i \ne x_j\}$. For a fixed real plane angle u put

$$D_u = T_n - R_u^* T_n. \tag{SA30}$$

First consider a configuration where both the original and rotated time coordinates are all distinct. In small product neighborhoods preserving both orders, approximate each smooth slot test in C^5 by finite sums $\alpha(x_4, x_1)\varphi(x_2, x_3)$. A product Fourier series with fixed compact cutoffs gives such approximants with support inside the same boxes and convergence through five derivatives. Equations (SA7) and (SA29) give equality for these plate tensors. Local C^5 continuity passes it to the original tests. Ordinary smooth tensor density then gives $D_u = 0$ as a distribution on that open set. Density of that open set alone would not exclude a singular part on its complement.

We need one local translation fact. If a strict coordinate-j cut separates a tuple into two groups with a positive gap, its moment after translating one group by a small real vector z has the Hilbert form

$$\langle \Phi, e^{-(d+z_j)H_j + iz_{\|}\cdot P_j}\Psi\rangle, \qquad d > 0. \tag{SA31}$$

This follows directly from the coordinate reflection and translation dictionary. For any real direction v, it is holomorphic in $z = sv$ on a neighborhood satisfying

$$d + v_j \operatorname{Re} s > \sum_{r\neq j} |v_r|\,|\operatorname{Im} s|. \tag{SA32}$$

The cube cone controls the exponential, and a smaller positive margin controls all its spectral derivatives. This is operator-norm holomorphy of translations, not a formal complex translation of a smooth test.

Fix any $y \in \mathcal{C}_n$. In each coordinate choose threshold cuts between successive distinct coordinate values of its points. These finitely many cuts separate every pair of indices. Choose a product neighborhood W and a small angle bound $u_0 < \pi/3$ such that, for $|u| < u_0$, each cut separates the SAME index groups for both x and $R_u x$, $x \in W$, with a uniform positive gap.

For fixed such u, let $q(x)$ be the number of original time ties plus the number of rotated time ties. The sets $\{q \le q_0\}$ are open. We prove by induction that D_u vanishes there. The case $q_0 = 0$ was just proved. At a point with $q = q_0 > 0$, take a tied pair in either frame and one of the fixed common cuts separating it. Translate the positive group of that cut by se_4. Original cross-group time differences have slopes ± 1; rotated ones have slopes $\pm\cos u$. These slopes are nonzero.

For a sufficiently small nonzero real s, all previously unequal time coordinates remain unequal and every cross-group tie becomes unequal. At least one tie disappears. Choose a small interval I of such s, then shrink a product neighborhood V of the original point so that its translated configurations all have at most $q_0 - 1$ ties for $s \in I$. The induction hypothesis makes D_u zero on those translated support boxes.

For arbitrary smooth slot tests in V, the translated difference is a holomorphic function of s near the real segment from zero to I. For its first term use (SA31)–(SA32) with direction e_4. For the rotated term use the same common coordinate cut and direction $R_u e_4$. The strict group gaps persist. It vanishes on I, so it vanishes at zero. Local product-tensor density proves that D_u vanishes as a distribution on V. The finite induction terminates at $q \le n(n-1)$.

This removes singular distributions supported on every exceptional time-tie set, including coincident planar projections with separated transverse coordinates. No analytic density across those projections was assumed.

69.1 1. A finite-energy transfer lemma

Let $H \geq 0$ be self-adjoint. Suppose ϕ is a translation-covariant Hermitian field, initially as a compatible family of forms, and that for $0 < s \leq 1$

$$A_s(f) = e^{-sH}\phi(f)e^{-sH} \quad \text{is bounded,} \qquad \|A_s(f)\| \leq Cs^{-q}\|f\|_{L^1(\mathbb{R}^4)}. \tag{EP1}$$

Compatibility means that these bounded forms arise from the same field distribution: inserting extra positive time translations on both sides gives the corresponding form at the larger time margin. This is provided by the analytic insertion-density construction, not inferred from separate bounds on unrelated operators.

Let f have compact Fourier support, and choose $R \geq 1$ so that the time-frequency projection of that support lies in $[-R, R]$. Write $P_n = \mathbf{1}_{[n,n+1)}(H)$, $n = 0, 1, \ldots$. Time-translation covariance and the Fourier support give

$$P_m A_s(f) P_n = 0 \quad \text{if} \quad |m - n| > R + 1. \tag{EP2}$$

For completeness, first smear the time translation of $A_s(f)$ against a Schwartz function whose Fourier transform separates the two energy intervals. The spectral theorem expresses the resulting matrix element as the Fourier transform evaluated at the energy difference. Translation covariance expresses the same quantity as the field evaluated on the corresponding convolution of f. If the two frequency

supports are disjoint, this convolution vanishes. Approximate the interval projections by bounded continuous spectral functions and pass strongly to the limit. This proves (EP2) without postulating boundedness of the unsandwiched field.

Choose $s_n = (1+n+R)^{-1}$. On the input band P_n, the output of $A_{s_n}(f)$ lies in energy at most $n+R+2$. Consequently both factors restoring the exponential weights are bounded on the relevant bands, by constants depending only on R. Define there

$$\phi(f)P_n = e^{s_n H}\mathbf{1}_{[0,n+R+2]}(H)A_{s_n}(f)e^{s_n H}P_n. \tag{EP3}$$

Compatibility makes this definition independent of the allowed positive value of s. Equation (EP1) gives

$$\|\phi(f)P_n\| \le C_R(1+n)^q\|f\|_{L^1}. \tag{EP4}$$

For $\psi \in \mathcal{H}$, set $v_n = \phi(f)P_n(1+H)^{-q}\psi$. Each v_n has energy support in $[n-R-1, n+R+2]$. These intervals have overlap multiplicity at most a fixed M_R. The spectral theorem and finite-dimensional Cauchy–Schwarz at each energy therefore imply

$$\left\|\sum_n v_n\right\|^2 \le M_R\sum_n \|v_n\|^2 \le C_R\|f\|_{L^1}^2\|\psi\|^2.$$

The series converges in norm, first on finite spectral sums and then on the whole Hilbert space. Thus

$$\boxed{\|\phi(f)(1+H)^{-q}\| \le C_R\|f\|_{L^1}.} \tag{EP5}$$

This is a one-sided bound. It is stronger than the original two-sided form bound for these compact-frequency tests. The finite energy-transfer property is indispensable; bounding each matrix block and then summing all output energies would not prove (EP5).

The actual-field domain needs an additional argument. Let $\mathcal{D}$ be the Wightman polynomial domain and let $\chi \in C_c^\infty(\mathbb{R})$. For $\psi \in \mathcal{D}$, write $\chi(H)\psi = \int g(t)e^{itH}\psi\,dt$, where g is Schwartz. Temperateness bounds $\|\phi(f)e^{itH}\psi\|$ by a polynomial in $|t|$. The integral therefore converges in the graph norm of the closure $\overline{\phi(f)}$: approximate it by truncated Riemann sums, each of whose vectors lies in $\mathcal{D}$. Its action is the integral of the corresponding field vectors. Translation covariance, compact energy transfer, and the compatible bounded forms identify that action with (EP3) on its compact input spectrum.

Choose $\chi = 1$ on $[0,E]$, with compact support in $(-1, E+1)$. For any $v \in \mathbf{1}_{[0,E]}(H)\mathcal{H}$, take $\psi_k \in \mathcal{D}$ tending to v. Then $\chi(H)\psi_k \to v$, and (EP4) on the fixed finite energy interval makes their field images Cauchy. Closedness implies $v \in \operatorname{Dom}\overline{\phi(f)}$, with action given by (EP3). Finally apply increasing spectral cutoffs to $(1+H)^{-q}\psi$. The square-sum estimate proving (EP5) makes their field images converge. Hence

$$(1+H)^{-q}\mathcal{H} \subset \operatorname{Dom}\overline{\phi(f)}, \qquad \|\overline{\phi(f)}(1+H)^{-q}\| \le C_R\|f\|_{L^1}. \tag{EP5a}$$

This identifies the bound with the closure of the reconstructed field. Agreement of forms on a merely dense set, without this graph-norm argument, would not establish the domain assertion.

69.2 2. The scalar plate bound supplies the hypothesis

Suppose the scalar angular and reconstruction arguments have produced a Wightman field, allowing a degenerate vacuum. The analytic insertion density of the preceding section represents $e^{-sH}\phi(0,\mathbf{x})e^{-sH}$ and obeys

$$\sup_{\mathbf{x}} \|e^{-sH}\phi(0,\mathbf{x})e^{-sH}\| \le Cs^{-q}, \qquad q = p + 9. \tag{EP6}$$

This notation denotes the already constructed bounded analytic density; it does not assert an unsmeared field operator before the positive time factors. Its identification with the reconstructed field follows by equality on real Euclidean insertions and uniqueness of the analytic matrix elements.

Real time translation acts by unitary conjugation and leaves the norm in (EP6) unchanged. Integrating that conjugate against a space-time test gives (EP1). With the small-ball source insertion bound proved in the history section one may take $p = 0$, so $q = 9$ is available. Larger fixed exponents would work equally well for (EP5).

Thus the scalar source construction supplies a generalized energy bound for every compact-frequency test, once the Euclidean covariance and Wightman reconstruction steps are justified. No mass gap is used.

69.3 3. The published pure-phase theorem and its exact role

Driessler and Summers, *On the decomposition of relativistic quantum field theories into pure phases*, Helvetica Physica Acta **59** (1986), 331–348, Theorem 2.3, prove a decomposition result for a Hermitian scalar Wightman field whose vacuum may be degenerate. One sufficient hypothesis is a generalized energy bound for every test with compact Fourier support. Its central decomposition then has, almost everywhere, Wightman components satisfying clustering and vacuum uniqueness. The source is https://doi.org/10.5169/seals-115699.

Equation (EP5) supplies this energy-bound hypothesis explicitly, with the strictly positive function $F_f(E) = (1 + E)^{-q}$. The theorem therefore offers a route to scalar vacuum selection after the other reconstruction properties have been proved. It does not create those properties, prove rotational covariance, or identify the components as Yang–Mills theories.

The decomposition also does not imply that every component retains a specified short-distance coefficient. Transferring such a coefficient to individual components requires a uniform or conditional ultraviolet estimate. That identification is a separate issue from the existence of a unique-vacuum component.

69.4 4. A nonconstant separated two-point function detects a nontrivial component

There is nevertheless a simple noncollapse consequence. If almost every pure component had only a constant scalar field $\phi_\xi(x) = c_\xi I$, then all its two-point functions would be constant in the separation. Their integral would have the same property. For fixed smooth nonnegative profiles f, h and their diameter-d rescalings of fixed amplitude, the connected two-point function would consequently be

$$\mathrm{Var}(c_\xi)\, d^8 \left(\int f\right)\left(\int h\right). \tag{EP7}$$

The variance is finite whenever the corresponding separated two-point moment is finite. Contact terms do not affect this calculation, because the two supports are separated.

An actual ultraviolet law of the form

$$\mathrm{Cov}(\phi(f_d),\phi(h_d)) = x_d^{-2}\{C(f,h)+o(1)\}, \qquad x_d \asymp \log(1/d), \qquad C(f,h)>0, \tag{EP8}$$

is incompatible with (EP7). Hence a positive-measure set of the Wightman components has a nonconstant scalar field. Equivalently, some field vector in those components has a nonzero projection outside the zero-energy subspace. This gives a nontrivial unique-vacuum scalar component when (EP8) is established for the physical theory. It does not establish an interacting gauge theory from a scalar two-point function alone.

69.5 5. A useful probability lemma, and why it is not silently substituted here

For an ordinary translation-invariant probability measure on a configuration space with a dense bounded local algebra, ergodic components inherit a reflection whenever the original measure is reflection invariant and reflection positive across all translates of that mirror. Here is the precise argument.

Let $\mathcal{I}$ be the translation-invariant sigma algebra. For $A \in L^2(\mathcal{I})$, choose bounded local approximants and translate each far enough into the positive half-space. Since A is invariant, the approximation still holds in L^2. Reflection positivity then yields $\langle \theta A, A\rangle \geq 0$. The reflection preserves $L^2(\mathcal{I})$ and is a unitary involution there. Positivity of this involution forces it to be the identity. Thus the reflection fixes the invariant sigma algebra pointwise, modulo null sets.

For a bounded positive-half local function F and a bounded nonnegative $h \in L^\infty(\mathcal{I})$, approximate $\sqrt{h}$ in L^2 by uniformly bounded positive-half local functions H_n. Apply reflection positivity to FH_n and pass to the limit. One obtains

$$\mathbb{E}\big[h\,\overline{\theta F}F\big] \geq 0.$$

Since h is arbitrary, $\mathbb{E}[\overline{\theta F}F \mid \mathcal{I}] \geq 0$ almost surely. A countable dense local algebra gives reflection positivity in almost every ergodic component on one common full measure set. Reflection invariance follows from the same conditional expectation identity. If the probability measure is fully Euclidean invariant, apply the argument first to a countable dense family of mirrors and then use continuity to obtain full Euclidean invariance of its components.

This lemma does not say that a limit of ergodic measures is ergodic. For example, stationary two-state Markov chains with flip probability $\epsilon > 0$ are ergodic, whereas their limit as $\epsilon \downarrow 0$ is the equal mixture of the two constant configurations. Nor does the lemma apply automatically to a contact-renormalized scalar Schwinger functional: reflection positivity does not make that functional a positive classical probability measure. The energy-bound route above addresses the reconstructed field directly and avoids that unsupported substitution.

70.1 1. An exact identity for mixtures of Gaussian scalar phases

Let a centered, Euclidean-invariant scalar hierarchy be a mixture of Euclidean-invariant Gaussian scalar hierarchies. The components may have different constant means and different two-point functions. Use the same smooth radial test profile at four distinct centers. In one component, let m be its smeared mean and let $c(s)$ be its centered two-point value when the squared center separation is s. The dependence on the common smearing scale is suppressed in this notation. Wick's rule gives

$$S_4 = m^4 + m^2 \sum_{i<j} c(s_{ij}) + c(s_{12})c(s_{34}) + c(s_{13})c(s_{24}) + c(s_{14})c(s_{23}). \tag{NG1}$$

The radial dependence follows from the component's connected Euclidean rotation covariance and the radial smearing, rather than from covariance of the mixture alone. Proper Poincaré covariance of each scalar phase supplies this connected rotation covariance after continuation. It already makes a scalar two-point function radial; Wick's rule then makes a Gaussian component's entire hierarchy invariant under all orthogonal transformations. No separate parity symmetry of arbitrary non-Gaussian phases is being inferred from symmetry of their mixture.

Now fix

$$s_{13} = s_{14} = s_{24} = s_{34} = 1, \qquad s_{12} = a, \quad s_{23} = b, \qquad a, b \in \{1, 2\}. \tag{NG2}$$

Each term in (NG1) is a function of a alone, a function of b alone, or a constant. Thus the mixed finite difference

$$\mathcal{D}F := F(2,2) - F(2,1) - F(1,2) + F(1,1) \tag{NG3}$$

annihilates S_4 in every Gaussian component. It also annihilates their mixture. There is no need to interchange a derivative with the phase integral; (NG3) is a finite linear combination of four already defined smeared moments.

The complete hierarchy is centered, so $\kappa_4 = S_4 - S_2(12)S_2(34) - S_2(13)S_2(24) - S_2(14)S_2(23)$. Each subtracted term again depends on one of a, b or neither. Consequently every such Gaussian mixture satisfies

$$\mathcal{D}\kappa_4 = 0. \tag{NG4}$$

This remains true with a different overall physical scale in (NG2), since all six squared lengths are then multiplied by the same factor. It also includes nonzero component means whose mixture has mean zero.

70.2 2. Four actual Euclidean configurations

The six distances in (NG2) are realizable simultaneously. Take $x_1 = 0$, and let x_2, x_3, x_4 have Gram matrix

$$G(a,b)=\begin{pmatrix} a & (a+1-b)/2 & a/2 \\ (a+1-b)/2 & 1 & 1/2 \\ a/2 & 1/2 & 1 \end{pmatrix}. \tag{NG5}$$

The first leading principal minor is $a>0$. At the four choices $(a,b)=(1,1),(1,2),(2,1),(2,2)$, the second leading minors are respectively $3/4,1,1,7/4$, and the determinants are $1/2,1/2,1/2,3/4$. Sylvester's criterion therefore gives a nondegenerate tetrahedron in $\mathbb{R}^3$, embedded in $\mathbb{R}^4$, for every choice. All pair distances are at least one. This avoids an impossible independent variation of distances in a degenerate collinear configuration.

70.3 3. The Maxwell-curvature quadratic violates the identity

For the free curvature quadratic $Q=\frac{1}{2}\sum_{a,I}:F_I^aF_I^a:$, the preceding higher-scalar calculation gives three connected four-cycle traces. Write d_G for the number of independent colors. All edge vectors of one tetrahedron lie in its three-dimensional span. A reflection in an edge-normal direction acts as the identity on the orthogonal fourth direction. The product of four reflections has determinant one on that span; hence

$$\mathrm{Tr}_{\Lambda^2\mathbb{R}^4}(R_1R_2R_3R_4)^{\wedge 2}=2\,\mathrm{Tr}_{\mathbb{R}^3}(R_1R_2R_3R_4). \tag{NG6}$$

In the basis x_2,x_3,x_4, an edge vector with coefficient column v has squared length $s=v^{\mathsf{T}}Gv$, and its reflection matrix is exactly

$$R_v=I-\frac{2v(v^{\mathsf{T}}G)}{s}. \tag{NG7}$$

These rational matrices give the following exact values. The common factor d_G/π^8 is omitted; a cycle is divided by the square of each of its four squared edge lengths, as prescribed by the curvature covariance $K(x)=\pi^{-2}|x|^{-4}R_x^{\wedge 2}$.

(a,b)	Cycle 12341	Cycle 12431	Cycle 13241	Total
$(1,1)$	0	0	0	0
$(1,2)$	$1/2$	-2	$1/2$	-1
$(2,1)$	$1/2$	$1/2$	-2	-1
$(2,2)$	$5/32$	$-1/4$	$-1/4$	$-11/32$

For example, each entry is obtained by taking the four differences of the columns $0,e_1,e_2,e_3$ around its indicated cycle, inserting them in (NG7), multiplying the resulting 3-by-3 matrices, and using (NG6). Thus the table is a finite rational calculation with the full normalization specified. It yields

$$\boxed{\mathcal{D}\kappa_4^Q=\frac{53\,d_G}{32\pi^8}\neq 0.} \tag{NG8}$$

To use honest tests instead of points, choose a nonnegative radial $\chi\in C_c^\infty(\mathbb{R}^4)$ of unit integral and put $\chi_\epsilon(x)=\epsilon^{-4}\chi(x/\epsilon)$. Place a copy at each vertex of each of the four tetrahedra. For sufficiently small ϵ, all supports remain separated. The Maxwell connected kernel is continuous on a neighborhood of

these finitely many configurations, so its smeared values converge to the table. Choose one fixed $\epsilon > 0$ small enough that their mixed difference is still positive, and then keep it fixed in the ultraviolet limit. No joint limit in the shape parameter and the physical cutoff is required.

70.4 4. Consequence of a physical four-point ultraviolet law

Suppose the same physical scalar continuum hierarchy has, for these four fixed families of tests scaled by a common d, with $f_{i,d}(x) = f_i(x/d)$. The fixed f_i already include the chosen centers and the unit-integral shape χ_ϵ. Thus the physical rescaling has fixed amplitude and integral d^4, rather than being a new unit-integral approximate identity. Assume

$$\kappa_4^\sigma(f_{1,d}, f_{2,d}, f_{3,d}, f_{4,d}) = g_d^8\{\kappa_4^Q(f_1, f_2, f_3, f_4) + o(1)\}, \qquad d \downarrow 0. \tag{NG9}$$

There are only four configurations, so an estimate uniform over this finite family suffices. Their common normalization g_d is the running coupling of the same physical sequence, not a separately tuned coupling for each shape. Equations (NG8)–(NG9), with the fixed smearing just chosen, imply $\mathcal{D}\kappa_4^\sigma \neq 0$ for all sufficiently small d. Equation (NG4) rules out a decomposition entirely into Gaussian scalar phases.

After the scalar Wightman and generalized energy-bound hypotheses of the pure-phase theorem are satisfied, almost every component is a Euclidean-continuable scalar Wightman theory with a unique vacuum. The separated Euclidean moments also decompose over these phases. To justify this assertion, first use compact-frequency finite-energy approximants to the fields in the direct-integral algebra. Their positive-time semigroups and sandwiched products decompose fiberwise. The actual bounded insertion densities are norm limits of these time-regularized approximants; their common plate bound dominates the finitely many ordered products. Passing to those limits gives the phase integral of the separated Euclidean moments. Thus the argument does not interchange an unbounded phase integral with analytic continuation without a bound.

The preceding contradiction shows that a set of positive decomposition measure consists of non-Gaussian scalar components. In particular, one can choose a non-Gaussian unique-vacuum scalar component. This improves the earlier conclusion that at least one component is nonconstant.

This conclusion still does not establish interacting Yang–Mills. The comparison field Q used to prove (NG8) is itself a non-Gaussian composite of free Maxwell fields. A full gauge theory requires the compatible invariant operator family and its dynamical identification. The result here closes a specific logical gap in scalar phase selection; it does not remove that remaining requirement.

71.1 1. A positive spectral weight

Fix $\mu > 0$ and $A > 0$, and define the positive measure

$$d\rho(s) = A\,\mathbf{1}_{s\ge\mu^2}\frac{s^2}{[\log(e + s/\mu^2)]^2}\,ds. \tag{ID1}$$

It grows at most polynomially. With the usual four-dimensional massive free scalar two-point function $\Delta_+^{(s)}$, set

$$W_2(x-y) = \int \Delta_+^{(s)}(x-y)\,d\rho(s). \tag{ID2}$$

This is a tempered positive-type distribution. Indeed its momentum measure, apart from the conventional Fourier constant, is

$$\mathbf{1}_{p_0>0}\mathbf{1}_{p^2\ge\mu^2}A\frac{(p^2)^2}{[\log(e + p^2/\mu^2)]^2}\,d^4p, \qquad p^2 = p_0^2 - |\mathbf{p}|^2. \tag{ID3}$$

The weight is positive, Lorentz invariant, supported in the forward cone, and bounded by a polynomial in the Euclidean norm of p. Restriction of Schwartz Fourier transforms to this measure therefore defines a continuous map into its L^2 space. Take symmetric Fock space over the closure of that map and use its creation and annihilation operators to define $\phi(f)$. The finite-particle Schwartz domain is invariant. The vacuum moments are the Wick pairings of (ID2), and every connected moment of order greater than two vanishes.

Locality follows by writing the commutator as the spectral integral of massive Pauli–Jordan distributions. Each vanishes at spacelike separation; the polynomial bound permits the integral after Schwartz smearing and preserves that support property. Poincare transformations act by their one-particle representation and its second quantization. Every one-particle energy is at least μ, so the vacuum is the unique zero-energy vector. These facts give the scalar Wightman properties directly. The mass threshold here is just a convenient choice for this example; no conclusion about a Yang–Mills mass gap is being made.

This is the standard generalized-free-field construction; see Dütsch and Rehren, *Generalized free fields and the AdS–CFT correspondence*, Section 2, https://arxiv.org/abs/math-ph/0209035. The logarithmic weight (ID1) and the following coefficient calculation are the specific example used here.

71.2 2. Its off-diagonal Euclidean covariance

For $r = |x| > 0$, the Euclidean covariance is the convergent integral

$$C(r) = \frac{A}{4\pi^2 r}\int_{\mu^2}^{\infty}\frac{s^{5/2}K_1(r\sqrt{s})}{[\log(e + s/\mu^2)]^2}\,ds. \tag{ID4}$$

The modified Bessel function K_1 is positive and decays exponentially at infinity. The positive-energy construction gives reflection positivity for separated Euclidean insertions. No claim that (ID4) is locally integrable at the origin is needed: the composite dimension in question requires the separated or diagonal-flat Euclidean test domain.

Put $u = \mu r$, $\ell = \log(1/u)$, and $t = r\sqrt{s}$. Then

$$C(r) = \frac{A}{2\pi^2 r^8} \int_u^\infty \frac{t^6 K_1(t)}{[\log(e + t^2/u^2)]^2}\, dt. \tag{ID5}$$

For each fixed $t > 0$, the logarithmic denominator divided by ℓ^2 tends to four. The limit may be passed under the integral with the following explicit bound. For $u \le t \le u^{1/2}$, use $K_1(t) \le C/t$ and a denominator at least one; the contribution after multiplication by ℓ^2 is at most $C\ell^2 u^3$, which tends to zero. For $t \ge u^{1/2}$, the denominator is at least ℓ^2, so the integrand after multiplication by ℓ^2 is bounded by the integrable function $t^6 K_1(t)$. Dominated convergence applies to this second region.

The required Bessel integral can be evaluated without a normalization convention left implicit. Using

$$K_1(t) = \frac{t}{4} \int_0^\infty v^{-2} e^{-v - t^2/(4v)}\, dv$$

and positive Fubini, first integrate $t^7 e^{-t^2/(4v)}$. Its integral is $768 v^4$. Hence

$$\int_0^\infty t^6 K_1(t)\, dt = 192 \int_0^\infty v^2 e^{-v}\, dv = 384. \tag{ID6}$$

It follows that

$$\boxed{C(r) = \frac{48A}{\pi^2} \frac{1 + o(1)}{r^8 [\log(1/(\mu r))]^2}} \qquad (r \downarrow 0). \tag{ID7}$$

The coefficient is any desired positive number after choosing A.

71.3 3. The same law on fixed separated shapes

Let f, h be smooth compact functions whose supports are separated by a positive distance, and set $f_d(x) = f(x/d)$, $h_d(x) = h(x/d)$. Their amplitudes are fixed while their support diameters and separation shrink proportionally to d. A change of variables and (ID7) give

$$\begin{aligned} S_2(f_d, h_d) &= d^8 \iint f(v) h(w) C(d|v - w|)\, dv\, dw \\ &= \frac{48A}{\pi^2 [\log(1/(\mu d))]^2} \left\{ \iint \frac{f(v) h(w)}{|v - w|^8}\, dv\, dw + o(1) \right\}. \end{aligned} \tag{ID8}$$

Uniformity of (ID7) over the compact interval of possible $|v - w|$ follows from the same domination proof, since that interval stays away from zero. If the profiles are nonzero and nonnegative, the coefficient in braces is strictly positive. Thus this generalized free field has precisely the kind of weakening two-point behavior used in the scalar angular argument.

It is nevertheless Gaussian as a scalar field. Its connected moments of orders three and above are identically zero, by its Fock construction. This example does not reproduce all the proposed Yang–Mills scalar moments, and it is not asserted to be a Wilson continuum limit. It disproves the inference from the scalar two-point law and scalar reconstruction properties alone to Yang–Mills identification.

71.4 4. What this changes in the proof target

A nonzero scalar ultraviolet coefficient can prove noncollapse of a scalar reconstructed sector and can enter the rotation argument. Its logarithm can record the running coupling used to define that observable. Identification of the complete Yang–Mills theory requires additional joint local information: compatible curvature observables, their mixing and short-distance relations, and the dynamical relations connecting them. A non-Gaussian composite in some other free theory would not settle that last issue either, as the earlier Wick-square calculation shows.

The existence construction should therefore retain these two separate claims: reconstruction of a nontrivial scalar sector, and construction and identification of the full invariant field hierarchy. Proving the first is useful progress, but it cannot be used as a definition that silently supplies the second.

72.4 4. Scalar energy bounds do not imply microscopic phase space

This verifies the proposed obstruction to using a scalar H-bound as an automatic route to Bostelmann's finite-dimensional point-field reconstruction.

72.4.1 4.1 One scalar field with infinitely many bounded masses

Take pairwise distinct masses m_n in a compact interval [m0,m1], with 0<m0<m1, and weights w_n>0 with sum w_n<infinity. On bosonic Fock space over

$$\mathfrak{h} = \bigoplus_n \mathfrak{h}_{m_n}$$

define the generalized free scalar field with Kallen–Lehmann measure

$$\mu = \sum_n w_n \delta_{m_n^2}.$$

For a spacetime test f, its one-particle vector has n-th component $\sqrt{w_n} f_{m_n}$. The sum of the weights and the compact mass interval give the usual tempered Gaussian field, a unique vacuum, and Gaussian all-order bounds. Moreover $H = d\Gamma(\omega) \geq m_0 N$, so creation/annihilation estimates imply

$$\|\phi(f)(1+H)^{-1/2}\| \leq C_f. \tag{FA5}$$

Thus this is a counterexample within scalar theories with ordinary polynomial energy bounds and completely controlled Gaussian moments.

72.4.2 4.2 The scalar local Weyl net contains isolated species

Let f be real with support in a fixed double cone O. Its mass-shell norm $c_f(m) = \|f_m\|^2$ is continuous and bounded for m in [m0,m1]. Polynomials are dense in L^2 of the finite measure $c_f(\sqrt{\lambda})\, d\mu(\lambda)$ on the compact mass-squared interval. Therefore for each n there are **real** polynomials p_k such that

$$p_k(\lambda) \longrightarrow \mathbf{1}_{\{m_n^2\}}(\lambda)/\sqrt{w_n}$$

in this weighted L^2 space. The tests $p_k(-\Box)f$ have the same support as f and their one-particle vectors converge to the isolated n-th species vector f_(m_n). Weyl operators are strongly continuous in their one-particle argument. Hence the local von Neumann algebra of the single scalar contains the Weyl operators for these isolated species vectors.

This argument uses polynomial density, not uncontrolled finite Lagrange interpolation on more and more mass points. It also works if one of the mass points is an accumulation point: the singleton indicator is an L^2 function for the atomic measure.

72.4.3 4.3 Failure of ordinary local phase-space compactness

Choose f so that its Fourier transform is nonzero uniformly on a fixed small momentum ball on every mass shell in [m0,m1]. A sufficiently small nonnegative spacetime bump has this property. Choose E>m1. After normalizing the isolated species vectors h_n to norm one,

$$\inf_n \|\mathbf{1}_{[0,E]}(\omega)h_n\| > 0.$$

With the Weyl convention $W(h)\Omega = e^{-\|h\|^2/2}e^{ia^*(h)}\Omega$, the one-particle projection of P_E W(h_n)Omega equals

$$ie^{-1/2}\mathbf{1}_{[0,E]}(\omega)h_n.$$

These vectors are mutually orthogonal and uniformly bounded below. The W(h_n) lie in the unit ball of A(O). The map $A \mapsto P_E A\Omega$ therefore has a non-precompact unit-ball image. Small weights do not cure this, because the local algebra permits arbitrarily large real multiples of the test before exponentiating to a norm-one Weyl operator.

72.4.4 4.4 Direct failure of Bostelmann's microscopic condition

The microscopic condition is a small-region finite-rank approximation condition; one should not rely solely on an unstated implication from it to the fixed-region compactness property. The same example defeats the microscopic condition directly.

In four dimensions, take $f_r(x) = f(x/r)$, supported in O_r, with integral f nonzero. Uniformly for m in [m0,m1], as r decreases to zero,

$$\|(f_r)_m\|^2 = r^6\{c_0 + o(1)\}, \qquad c_0 > 0,$$
$$\|\mathbf{1}_{[0,E]}(\omega_m)(f_r)_m\|^2 \geq cr^8.$$

The first assertion follows by the change of spatial momentum k=r p in the mass-shell integral and dominated convergence to mass zero. For the second, integrate over one fixed momentum ball below E and use $\widehat{f}(r\omega, rp) \to \widehat{f}(0)$. Normalize the isolated species vectors h_(n,r) to norm one. Their fixed-low-energy norms obey

$$\|\mathbf{1}_{[0,E]}(\omega)h_{n,r}\| \geq cr. \tag{FA6}$$

uniformly in n and small r.

For any finite-rank map psi in Bostelmann's space Psi, the map obtained by applying its dual to A, then applying the resulting energy-damped form to Omega and projecting onto the one-particle energies <=E, has finite-dimensional range. The actual local Weyl outputs have infinitely many orthogonal one-particle components of norm at least c r. For each fixed r, their distance from any fixed finite-dimensional range is therefore at least c r in the supremum limit over n. Testing with the associated finite-energy vector functionals yields

$$\|\Xi - \psi\|_r^{(\ell)} \geq c_\ell r. \tag{FA7}$$

for every finite rank psi and every finite ell for which its seminorm is defined. The energy damping only changes the constant by a power of 1+E. Consequently delta_gamma(Xi-psi) cannot vanish at gamma=1, or at any larger gamma. The microscopic phase-space condition fails.

The relevant primary definitions are Bostelmann, *Phase space properties and the short distance structure in quantum field theory*, arXiv:math-ph/0409070, Section II, and *Operator product expansions as a consequence of phase space properties*, arXiv:math-ph/0502004, equations (2.5)–(2.7). The counterexample proves that an actual finite-rank short-distance estimate would be an additional Wilson input; it cannot be inferred from the scalar energy bound, Wightman axioms, or Gaussian all-order growth alone.

73.1 1. Conditioning and a genuine observation buffer

Let X be a centered Gaussian vector on a finite-dimensional real space with strictly positive covariance C. Let Q be a surjective coarse observation, and let T be a full-row-rank map selecting the variables on which a bounded measurable observable depends. Write

$$H = CQ^*(QCQ^*)^{-1}, \qquad \Gamma = C - CQ^*(QCQ^*)^{-1}QC, \qquad M = TH, \quad \Sigma = T\Gamma T^*. \tag{BA1}$$

Conditionally on $QX = v$, the vector TX has mean Mv and covariance Σ.

There is an exact nondegeneracy criterion:

$$\ker \Sigma = \{w : T^*w \in \operatorname{ran} Q^*\}, \qquad \Sigma > 0 \iff \operatorname{ran} T^* \cap \operatorname{ran} Q^* = \{0\}. \tag{BA2}$$

To prove it, put $B = QC^{1/2}$ and let $P = B^*(BB^*)^{-1}B$. Then $\Gamma = C^{1/2}(I - P)C^{1/2}$, so its kernel is exactly $\operatorname{ran} Q^*$. Taking the quadratic form on T^*w proves (BA2), using the injectivity of T^* for the last equivalence.

For disjoint scalar block averages, a strict observation buffer is one simple way to obtain this algebraic condition. If every block touched by the support of T also has at least one unobserved site, a vector constant on each block and supported entirely in the observed sites must vanish. Thus the ranges in (BA2) are disjoint. This assertion concerns independent scalar block averages; a gauge-constrained or overlapping map requires checking its own ranges.

Nondegeneracy alone gives no uniform lower eigenvalue bound. Both the size of the buffer and the covariance matter in quantitative estimates. If $T = Q$, there is no buffer: $\Sigma = 0$, and the conditional expectation of a bounded discontinuous function of QX is that same discontinuous function of v. Conditional expectation is therefore not automatically an analytic smoothing operation.

73.2 2. Entire dependence without evaluating an observable at complex fields

Assume $\Sigma > 0$, and let A be a bounded measurable function of the observation vector. Define, for $z \in \mathbb{C}^d$,

$$F_A(z) = \frac{1}{(2\pi)^{d/2}\det(\Sigma)^{1/2}} \int_{\mathbb{R}^d} A(y) \exp\left[-\frac{1}{2}(y - z)^T\Sigma^{-1}(y - z)\right] dy. \tag{BA3}$$

The exponent uses the bilinear transpose; it does not conjugate z. The integral is entire by domination on each compact complex set. For $z = m + iu$ with m, u real, its absolute value obeys

$$\boxed{|F_A(m + iu)| \le \|A\|_\infty \exp\left(\frac{1}{2}u^T\Sigma^{-1}u\right).} \tag{BA4}$$

Indeed the real part of the exponent equals $-\frac{1}{2}(y-m)^T\Sigma^{-1}(y-m) + \frac{1}{2}u^T\Sigma^{-1}u$, and the first Gaussian integrates to one. In particular

$$\mathbb{E}[A(TX) \mid QX = v] = F_A(Mv) \tag{BA5}$$

for real v, with the indicated entire continuation. Only the density in (BA3) was continued. The observable A itself remains evaluated at real y; it need not possess a holomorphic extension, derivatives, or even continuity.

This distinction matters for the proposed bounded-observable route. A complex background change inside a formula $A(Hv+\zeta)$ cannot be justified by calling A bounded. A change of integration variables that puts the complex dependence in a uniformly controlled density can justify it, as (BA3) does.

73.3 3. An exact derivative estimate

Use the Cameron–Martin norm $\|u\|_\Sigma = (u^T\Sigma^{-1}u)^{1/2}$. For real $m, u_1, \dots, u_k$, differentiation of the Gaussian density gives

$$D^k F_A(m)[u_1,\dots,u_k] = \mathbb{E}\left[A(m+Z) : \prod_{j=1}^{k} u_j^T\Sigma^{-1}Z :\right], \qquad Z\sim\gamma_\Sigma. \tag{BA6}$$

The Wick ordering in (BA6) is with respect to the joint covariance of the displayed linear Gaussian variables. Its second moment is the permanent of their Gram matrix. Each term of that permanent has absolute value at most $\prod_j \|u_j\|_\Sigma^2$, by Cauchy–Schwarz. There are $k!$ terms. A second Cauchy–Schwarz inequality therefore gives

$$\boxed{|D^k F_A(m)[u_1,\dots,u_k]| \le \sqrt{k!}\,\|A\|_\infty \prod_{j=1}^{k}\|u_j\|_\Sigma.} \tag{BA7}$$

This bound is independent of the observation dimension. All possible cutoff dependence is explicit in the Cameron–Martin norms of the shift directions. It cannot be discarded by treating Σ as a fixed covariance when the fine lattice changes.

For $k=1$, integrate (BA7) along a real segment to obtain

$$|F_A(m)-F_A(m')| \le \|A\|_\infty \|m-m'\|_\Sigma. \tag{BA8}$$

For repeated direction m, Taylor's integral remainder gives the useful finite-jet estimate

$$\left|F_A(m) - \sum_{k=0}^{p}\frac{1}{k!}D^kF_A(0)[m^{\otimes k}]\right| \le \|A\|_\infty \frac{\|m\|_\Sigma^{p+1}}{\sqrt{(p+1)!}}. \tag{BA9}$$

These are exact nonperturbative finite-Gaussian estimates. No power series in a Yang–Mills coupling is involved.

73.4 4. A finite-rank approximation consequence

Suppose the admissible real coarse data form a set $\mathcal{V}$, and a linear map S of rank at most d_0 approximates M in the following sense:

$$\sup_{v\in\mathcal{V}}\|(M-S)v\|_\Sigma \le \epsilon, \qquad \sup_{v\in\mathcal{V}}\|Sv\|_\Sigma \le R. \tag{BA10}$$

The map from A to the function

$$(\mathcal{P}_pA)(v) = \sum_{k=0}^{p}\frac{1}{k!}D^kF_A(0)[(Sv)^{\otimes k}] \tag{BA11}$$

has range dimension at most $\binom{d_0+p}{p}$. Its coefficients are bounded linear functionals of A, and its range consists of polynomials in d_0 fixed coarse coordinates. Combining (BA8) and (BA9) proves

$$\sup_{v\in\mathcal{V}} |F_A(Mv) - (\mathcal{P}_p A)(v)| \le \|A\|_\infty \left(\epsilon + \frac{R^{p+1}}{\sqrt{(p+1)!}}\right). \tag{BA12}$$

For a shrinking physical observation region of radius r, this gives a power-law approximation if, for example, a rank bounded independently of the fine cutoff and r supplies

$$\epsilon_r \le C r^{\gamma+\eta}, \qquad R_r \le C r^{\alpha}, \qquad \alpha, \eta > 0, \tag{BA13}$$

and one chooses a fixed integer p with $\alpha(p+1) > \gamma$. The rank may depend on the desired accuracy γ; it must not grow when r is subsequently taken to zero. When R_r stays of order one, making p grow with $\log(1/r)$ is not a finite-rank proof of this kind.

There is also a compatibility requirement. Approximations with different ranges at every r do not by themselves establish a common point-field space. A useful elementary observation is the following. If all approximants lie in one fixed finite-dimensional subspace V of a normed space, and P is a bounded projection onto V, then

$$\|x - Px\| \le (1 + \|P\|) \inf_{v\in V} \|x - v\|. \tag{BA14}$$

Thus a common range can replace scale-dependent coefficient formulas by one fixed projection. In a quantum phase-space argument, the corresponding fields, energy-damped norms, and normal functional coefficients must still be constructed; (BA14) alone does not identify them.

74.1 1. A conditional scalar construction theorem

Fix one bare Wilson sequence, with the physical scale convention and coupling shooting chosen once. Suppose its centered scalar action insertions have the following properties along compatible periodic volumes.

Separated distribution control. Their moments satisfy the all-cutoff separated-cube bound (D1), with $s = 5, b = 5, p = 9$, and the corresponding bounds under small translations of fixed separated supports. The constants have the fixed dependence on the number of insertions required there. The physical coordinate and diagonal reflection forms are nonnegative and their finite support collars vanish with the cutoff.

Uniform one-insertion strength. The original physical source functional has the ordinary source bounds and strictly local contact normalizer used in the chessboard argument. Its raw polynomial reflection estimate is uniform in every surrounding polynomial, as in the marked-history insertion theorem. After passage to a common limit it gives (SP2), with $s = 5, p = 0$. Bounds for each fixed correlation without this uniformity are insufficient.

Physical ultraviolet matching. On the same sequence, the fixed-reference two-point comparison, including all complementary histories and retained-field terms, holds uniformly over the fixed positive shape class used in the two-point angular proof. The volume transfer uses its common source normalizer. For at least one pair of fixed nonnegative profiles the limiting law has the form

$$S_2(f_d, h_d) = x_d^{-2}\{C(f,h) + o(1)\}, \qquad C(f,h) > 0, \qquad x_d \asymp \log(1/d). \tag{SC1}$$

If the four-label comparison is also used, it is on this same sequence and with its same x_d.

Under these hypotheses a subsequence reconstructs a nontrivial scalar Wightman theory. Its central decomposition contains a nonconstant component with a unique vacuum. If in addition the physical four-point comparison holds for the four fixed radial-bump families of the preceding section, the decomposition contains a non-Gaussian component with a unique vacuum.

The words “under these hypotheses” are essential. The preceding chapters give detailed derivations and actual-map interfaces for them. This theorem states precisely what their simultaneous application would prove; it does not independently certify every model-specific estimate by restating it as a hypothesis.

74.2 2. Proof of the implication

At each fixed mesh first take a compatible periodic-volume subsequence. The uniform source and common-normalizer estimates pass to that limit on bounded finite-cutoff observables. Compact link-variable probability measures provide the fixed-mesh subsequence; they do not provide continuum field regularity.

The all-cutoff product-test bound and the explicit dyadic flat-test extension give

$$|S_{n,a}(F)| \le AB^n(n!)^p p_{kn}(F), \qquad F \in \mathcal{S}_{\Delta,n}, \tag{SC2}$$

with fixed finite p, k. Choose countable dense test sets for every n, and extract one diagonal continuum subsequence. The same uniform seminorm extends convergence from those tests to every flat test and to separated compact tensors. This is one hierarchy, not a different subsequence for each correlation.

Finite Gram matrices of the physical reflection form converge entrywise and remain nonnegative. Hypercubic symmetry passes directly. Ordinary translations pass by approximating each real translation with lattice translations and using the uniform local test continuity. In each coordinate Hilbert space the resulting time semigroup and spatial translation group are strongly continuous. The earlier diagonal-reflection spectral carrier argument supplies the cube cone $|P_{j,r}| \le H_j$; it is not inferred from coordinate reflection positivity alone.

The raw one-insertion inequality passes in its finite polynomial matrix elements and their norm squares. Its null-space property therefore survives in the limit. The compact Δ^5 parametrix gives bounded analytic point insertions with time-margin exponent nine. The independent two-point argument uses (SC1)'s upper bound on the prescribed shapes to obtain $t^8\Gamma(t) \to 0$, and then uses the positive diagonal spectral measure to remove its last nonconstant harmonics.

The actual boosted product vectors, finite angular band, polygon divisibility and integer convexity consequently remove every higher scalar angular harmonic. The common-cut analytic translation argument removes possible distributions on exceptional projection sets. Thus the entire hierarchy, including its flat-test extension, is fully Euclidean covariant.

The corrected Osterwalder–Schrader reconstruction uses (SC2), covariance, symmetry and reflection positivity. Before a clustering property is imposed, the vacuum is allowed to be degenerate. This is the no-clustering subset of Osterwalder–Schrader reconstruction: see OS II, Sections II and IV.1–2 (printed pp. 283–284 and 287–288), its introductory Remark 6, and OS I, Remark 3 (printed p. 89), which explicitly records the corresponding subset implication. The time-regularized scalar insertion represents the reconstructed field by analytic uniqueness and gives the compatible energy bound described in (EP1)–(EP6). For tests with compact energy-frequency support, the finite-energy-transfer argument proves the actual-field domain inclusion and

$$\|\overline{\phi(f)}(1+H)^{-9}\| \le C_f < \infty. \tag{SC3}$$

The graph-norm closure step in that argument is part of the conclusion; equality of forms on an unspecified dense set is not being substituted for it.

The Driessler–Summers pure-phase theorem (Theorem 2.3, printed p. 335, with its energy-bound and direct-integral arguments on pp. 337–343) now applies to the scalar field. Almost every component has proper Poincaré covariance; connected Euclidean rotation covariance follows for its separated continued moments. The bounded time-regularized insertions and finite-energy approximants justify their phase integral as explained in the four-shape section. Equation (SC1) rules out a decomposition entirely into constant fields, since such a mixture would have a connected two-point function proportional to d^8 on amplitude-one shrinking profiles. Hence a nonconstant unique-vacuum component exists. With the additional four-point law, (NG4)–(NG9) exclude a decomposition entirely into Gaussian scalar components. This proves the asserted stronger conclusion. No mass gap is used in any of these steps.

74.3 3. Why this is still short of the requested proof

The theorem constructs only the action-density cyclic sector. The dimension-four bilinear and dimension-six charge-odd calculations now include a conditional joint construction: exact subset

transport, retained lower-degree outputs, fixed-order and Gevrey separated bounds, and a local replicated-cell estimate through the represented large-field state. These results retain the specified full-core local-potential and original history hypotheses. They do not construct a complete common physical field hierarchy. In particular, general tensor transformation laws and simultaneous reconstruction cannot be supplied by the scalar angular theorem.

The cubic first-insertion statement is specifically for $SU(N)$, $N \geq 3$. Charge conjugation protects the lower jets of a single odd cubic insertion; two such insertions have even parity and can generate the identity, curvature-square, and other lower-dimensional mixed coefficients. Their joint transport therefore requires the additional subset and history estimates in the joint-curvature chapters, even though the one-insertion normalization is a finite invertible cocycle. Those estimates do not remove the derivative filtration needed for higher first insertions.

For general invariant polynomials involving arbitrarily many covariant derivatives, the sharp conditional averages exhibit the explicit boundary singularity proved in the higher-jets section. The smooth-cylinder tilt theorem handles a genuine nonlinear class, but its constants are not a uniform estimate for the ultraviolet-local Wilson interaction. A construction of that full operator family needs a cancellation-aware completed coefficient estimate for near-mark interaction attachments, or a different observable/blocking construction with a proved comparison for the same bare sequence.

Finally, non-Gaussian scalar fields include composites of free fields. The four-shape result removes a Gaussian-mixture loophole in phase selection; it does not identify the complete dynamics as interacting Yang–Mills. The required joint local relations, short-distance normalization and dimension-filtered operator correspondence remain part of the proof target.